

Professor: Prof. Massimiliano Marzo

Portfolio Theory

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Group 8

Alessandro Bonfiglio —

Atabak Nikouseresht —

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1 Q1&Q2—Descriptive Statistics: Nasdaq and NYSE Securities

This section presents a combined, unified analysis of the descriptive statistics for Nasdaq and NYSE securities using daily and monthly returns computed from adjusted prices:

$$R_{i,t} = \frac{P_{i,t}}{P_{i,t-1}} - 1$$

where $R_{i,t}$ is the return of security i at time t , $P_{i,t}$ is the adjusted price at time t , and $P_{i,t-1}$ is the previous period price.

There are 1305 observations for the daily data set and 60 for the monthly data set, whilst the number of securities used in the analysis is 131 for the NYSE and 80 for the Nasdaq. For each security, the following descriptive metrics are computed: mean, standard deviation, variance, skewness, and excess kurtosis (Fisher definition, excess over 0 for normal distribution).

Security	Daily Mean	Daily Std Dev	Daily Variance	Daily Skewness	Daily Exc. Kurtosis	Monthly Mean	Monthly Std Dev	Monthly Variance	Monthly Skewness	Monthly Exc. Kurtosis
APPLE	0,0008	0,016954	0,000287	0,478185	7,386054	0,017169	0,075986	0,005774	0,136048	-0,61239
NVIDIA	0,002622	0,031931	0,00102	0,552982	5,016527	0,057449	0,152831	0,023357	0,865475	1,350047
MICROSOFT	0,00054	0,016299	0,000266	0,029573	4,157087	0,010903	0,064389	0,004146	0,132208	0,462105
AMAZON	0,000618	0,021879	0,000479	0,137653	5,3099	0,01232	0,090048	0,008109	-0,25463	0,962585
META PLATFORMS	0,000881	0,027112	0,000735	-0,16158	21,15996	0,017903	0,116386	0,013546	-0,14135	0,850448
TESLA	0,001245	0,036395	0,001325	0,301244	3,375157	0,030396	0,201441	0,040579	1,224829	2,235019
ALPHABET 'A'	0,001142	0,019293	0,000372	0,187788	3,664448	0,023664	0,074056	0,005484	0,04167	0,413658
BROADCOM	0,00208	0,026384	0,000696	0,918753	10,53822	0,043846	0,109882	0,012074	1,060973	1,582909
COSTCO WHOLESALE	0,000867	0,013934	0,000194	-0,63501	8,26709	0,019408	0,071695	0,00514	-0,36945	0,171381
NETFLIX	0,000804	0,026608	0,000708	-1,63467	29,78064	0,020051	0,144584	0,020904	-0,03249	1,2566
T-MOBILE US	0,000314	0,014697	0,000216	-0,12458	6,970679	0,006974	0,072368	0,005237	0,858705	1,751885

Table 1: Descriptive Statistics: Daily and Monthly (Nasdaq Securities)

Security	Daily Mean	Daily Std Dev	Daily Variance	Daily Skewness	Daily Exc. Kurtosis	Monthly Mean	Monthly Std Dev	Monthly Variance	Monthly Skewness	Monthly Exc. Kurtosis
3M	3,08E-05	0,017479	0,000306	1,70468	26,83532	0,000776	0,082575	0,006819	0,095521	0,913248
AAR	0,000975	0,021701	0,000471	-0,22281	6,299124	0,021733	0,107004	0,01145	0,072354	-0,52757
ABBOTT LABORAT.	-0,00016	0,013796	0,00019	-0,66451	6,350844	-0,00391	0,058631	0,003438	0,204244	0,882024
AMERICAN EXP.	0,000694	0,018178	0,00033	0,124704	5,855573	0,015007	0,084099	0,007073	0,078587	-0,7112
BANK OF AMER.	0,000261	0,016589	0,000275	-0,0258	3,627339	0,006176	0,084217	0,007093	0,035801	-0,24466
BK OF NEW YORK	0,00085	0,015195	0,000231	-0,20803	3,826297	0,018945	0,075654	0,005724	-0,46343	0,038412
BARRICK MINING	0,000649	0,022213	0,000493	0,014538	2,872034	0,014428	0,106936	0,011435	-0,11241	-0,79758
BOEING	0,000226	0,0225	0,000506	0,046326	3,95892	0,006687	0,123071	0,015147	0,456726	0,478728
BRISTOL MYERS S.	1,03E-05	0,014829	0,00022	0,459898	6,931219	-0,00013	0,065012	0,004227	0,835373	1,648264
CABOT	0,000411	0,020613	0,000425	0,869397	6,766314	0,007312	0,077814	0,006055	0,337139	-0,45045
CALERES	-6,3E-05	0,034435	0,001186	0,450519	8,364172	-0,00386	0,144996	0,021024	0,38592	0,156408

Table 2: Descriptive Statistics: Daily and Monthly (NYSE Securities)

Several observations follow: first, daily returns are smaller than monthly returns due to shorter observation windows. Monthly returns aggregate price movements, resulting in higher mean values in both markets.

Talking about volatility, Nasdaq exhibits higher standard deviations and variances compared to NYSE, reflecting its technology and growth-oriented composition. Highly volatile Nasdaq stocks include Quantum Computing, Rocket Lab, Coinbase Global, and AppLovin. NYSE shows lower volatility overall, but certain industrial and biotech stocks (e.g., Foot Locker, TEAM, Biohaven) are an exceptions to this fact.

Moving on to skewness, Nasdaq stocks display mixed behavior, with positive skewness indicating potential for extreme positive returns, and negative skewness indicating downside risks. In this case too, NYSE skewness is generally less extreme but still highlights asymmetric return distributions in specific securities.

Finally, the last aspect analyzed is the excess kurtosis. In this case, it's possible to see that both markets present the same pattern. Nasdaq stocks such as ASTROTECH (182.95) and Quantum Computing (24.48) demonstrate extremely fat tails, indicating frequent large deviations from the mean, while NYSE also exhibits high kurtosis in selected securities like TELESYSTEMS & DATA (209.63). This shows extremely fat tails, highlighting potential for extreme events in risk assessment.

These statistics provide a comprehensive view of the risk-return characteristics of the two markets. Nasdaq's higher volatility and kurtosis suggest greater portfolio risk and need for diversification, while NYSE's broader sector diversification moderates risk. Both daily and monthly perspectives are crucial: daily data captures short-term fluctuations, whereas monthly aggregation reflects longer-term performance and accumulated risk, guiding portfolio selection and construction strategies.

2 Q3 — Covariance and Correlation Structure Analysis

After capturing the marginal risk of each security with descriptive statistics, we move on analyzing the joint behavior of returns, using covariance and correlation. This section examines the dependence structure of Nasdaq and NYSE securities using both daily and monthly return data. The covariance between two assets i and j is defined as:

$$Cov(R_i, R_j) = \frac{1}{T-1} \sum_{t=1}^T (R_{i,t} - \bar{R}_i)(R_{j,t} - \bar{R}_j)$$

where $R_{i,t}$ and $R_{j,t}$ are returns at time t , $\bar{R}_i$ and $\bar{R}_j$ are average returns, and T is the number of observations. Covariance captures the direction and magnitude of co-movement but is sensitive to the scale of individual asset volatilities, making cross-asset comparisons difficult. The correlation coefficient addresses this limitation by standardizing covariance:

$$\rho_{ij} = \frac{Cov(R_i, R_j)}{\sigma_i \sigma_j}$$

where σ_i and σ_j are the standard deviations of the securities.

Correlation analysis is essential in portfolio theory, as diversification benefits depend not only on individual asset volatility but also on how asset returns move relative to one another. Securities with lower correlations contribute more effectively to risk reduction. To support the asset selection process in Questions 4 and 5, the average absolute correlation of each security is also computed:

$$\bar{\rho}_i = \frac{1}{N} \sum_{j=1}^N |\rho_{ij}|$$

where N is the total number of securities.

This measure summarizes how strongly each security co-moves with the rest of the market on average, and serves as the primary diversification criterion for portfolio construction.

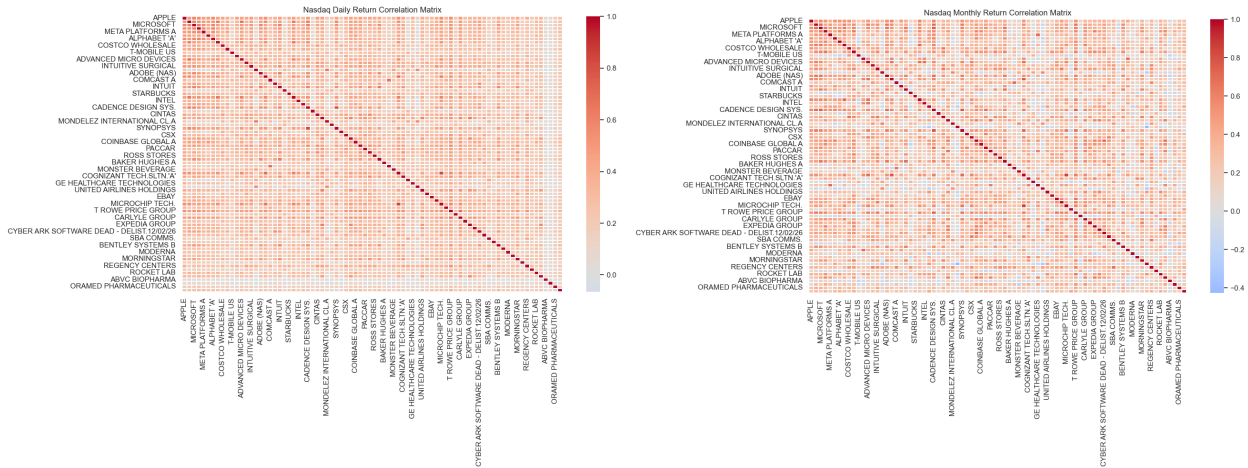


Figure 1: Daily and monthly correlation matrices for Nasdaq securities.

The Nasdaq correlation matrices reveal a clustering pattern among large-cap technology and semiconductor firms. Companies such as Nvidia, AMD, Broadcom, Marvell Technology, AppLovin, Coinbase, and Palo Alto Networks display strong positive pairwise correlations.

At the other end of the spectrum, more defensive and less cyclical names such as PepsiCo, Kraft Heinz, Gilead Sciences, and Exelon show noticeably lower, and in some cases slightly negative, correlations with the technology cluster. This makes them natural candidates for a diversified Nasdaq portfolio.

A recurring empirical pattern worth highlighting is that monthly correlations are consistently stronger and more dispersed than daily correlations across all Nasdaq securities. Monthly aggregation smooths out this noise and allows the longer-term economic relationships between assets to emerge more clearly.

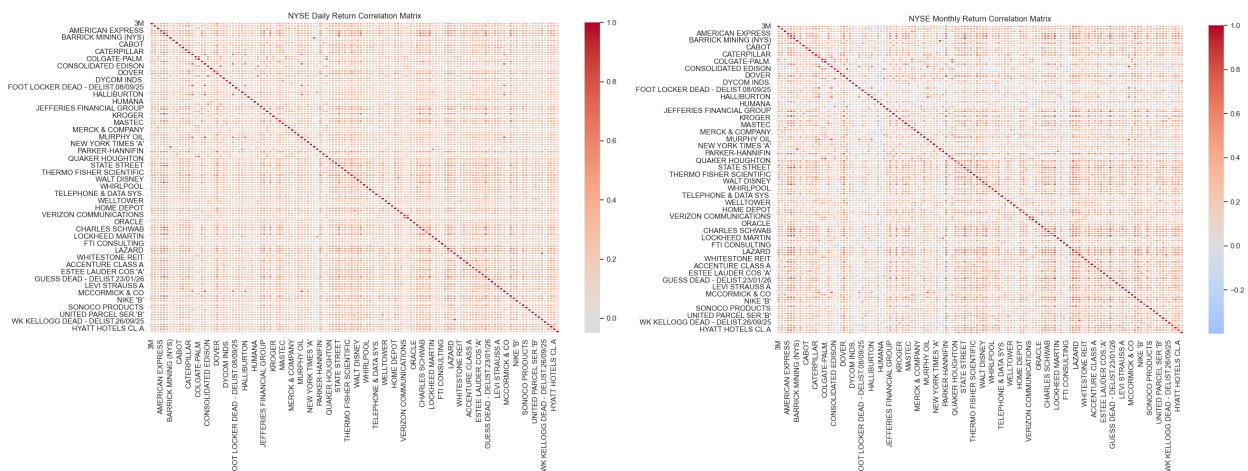


Figure 2: Daily and monthly correlation matrices for NYSE securities.

Compared with Nasdaq, the NYSE universe is broadly diversified across financial, industrial, consumer, energy, utility, telecom, and healthcare sectors. This sectoral breadth naturally limits the degree of correlation clustering visible in the heatmaps.

Defensive names such as Coca-Cola, Duke Energy, Verizon, and Colgate-Palmolive show weaker correlations with cyclical and growth-oriented securities, reflecting the well-known tendency of consumer staples and utilities to decouple from the broader market during periods of risk-off sentiment.

There are few differences between the two markets. Nasdaq exhibits a more tightly clustered correlation structure, driven by the dominance of technology-related firms; while NYSE, by contrast, benefits from broader sectoral coverage.

Technically, it is worth noting that both variance-covariance matrices are computed using the standard sample estimator with $T - 1$ degrees of freedom, ensuring unbiasedness.

3 Q4&Q5 — Selection of Nasdaq and NYSE Securities

The selection methodology adopted here builds directly on the correlation analysis of Q3. For each security, the average absolute correlation with respect to the full universe of available assets is computed as previously mentioned, and then are chosen the ten candidates with the lowest average absolute correlation, using a data-quality filter that excludes any security with more than 5% missing observations across the daily return sample to improve the reliability of the portfolio optimization process. The rationale for using average absolute correlation rather than, say, average return or volatility as the primary selection criterion is straightforward: diversification benefits in a portfolio context are driven by low co-movement, not by individual asset characteristics. A high-return asset that moves in lockstep with all other selected assets adds very little to portfolio efficiency. A lower-return asset that moves independently can meaningfully reduce portfolio variance and improve the Sharpe Ratio of the resulting allocation.

For the Nasdaq, the ten securities selected are: ABVC Biopharma, Astrotech, Kraft Heinz, Quantum Computing, CME Group, Oramed Pharmaceuticals, Exelon, Gilead Sciences, PepsiCo, and Regeneron Pharmaceuticals. What stands out immediately about this list is how different these companies are from each other and from the large-cap technology names that dominate the Nasdaq index. As shown in Q3, firms like Nvidia, AMD, Broadcom, and AppLovin form a tightly correlated technology cluster. Including any of them in the portfolio would essentially mean taking on the same underlying risk factor multiple times.

Taken together, the selected Nasdaq portfolio spans pharmaceuticals, biotechnology, speculative technology, financial exchanges, regulated utilities, consumer staples, and large-cap healthcare. The ten securities selected from the NYSE universe are: TEAM, Kroger, FTI Consulting, McCormick & Co, Nordic American Tankers, Humana, Telephone & Data Systems, Eli Lilly, Northrop Grumman, and Newmont.

Compared with the Nasdaq selection, the NYSE portfolio skews more heavily toward mature, as a consequence of NYSE's broader composition, which exhibits lower average pairwise correlations than Nasdaq. The selected NYSE portfolio covers software, consumer staples, professional services, shipping, healthcare insurance, telecommunications, pharmaceuticals,

defense, and precious metals.

The difference between two portfolios is that Nasdaq one is more heterogeneous in terms of individual asset risk, having speculative names Astrotech and Quantum Computing and defensive securities like Exelon, Kraft Heinz, PepsiCo; while NYSE presents less extreme variation in individual volatility profiles.

What they have in common is the same basic logic: to avoid the correlated core of each market and build a structure driven by different economic forces.

4 Q6 — Price Behavior of Selected Securities

This section analyzes the price behavior of the selected Nasdaq and NYSE securities using both daily and monthly observations, to show the full trajectory, including periods of sustained growth, sharp reversals, quiet consolidation, and extreme events. Since the selected securities have different original price levels, all price series are normalized to a common base value of 100 at the beginning of the sample period:

$$P_{i,T}^* = \frac{P_{i,t}}{P_{i,0}} \cdot 100$$

where $P_{i,T}^*$ is the normalized price index of security i at time t , $P_{i,t}$ is the actual adjusted price, and $P_{i,0}$ is the initial price.

Daily and monthly plots are produced separately. The daily charts capture the full granularity of short-term price movements while monthly charts filter out this noise and highlight the slower-moving trends that tend to be more relevant for medium-term portfolio decisions.

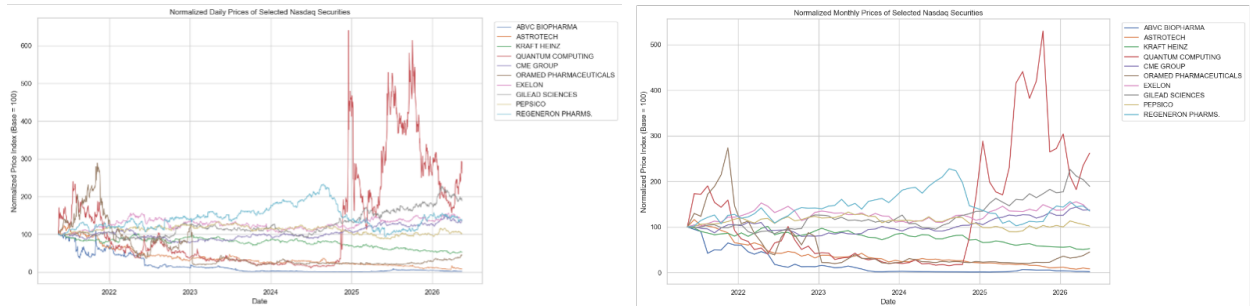


Figure 3: Normalized price dynamics of selected Nasdaq securities using daily and monthly data.

The price dynamics for selected Nasdaq securities shows what anticipated in previous question (Figure 3). The ten securities do not move together but they diverge substantially over the sample period, which is exactly what we are looking for.

It is worth taking a look at the behaviour of Quantum Computing and Astrotech: both exhibit sharp, nearly vertical spikes toward the end of the sample period followed by partial reversals, a pattern of speculative micro-cap and early-stage technology securities, prices can remain dormant for extended periods and then surge on a single event before mean-reverting as the initial excitement fades.

At the other end of the spectrum, Kraft Heinz, PepsiCo and Exelon display gradual, relatively smooth trajectories that remain close to or slightly below their starting value, consistent with the defensive nature of consumer staples and regulated utilities, and their presence in the portfolio is useful to counterbalance the extreme volatility of the two securities mentioned above.

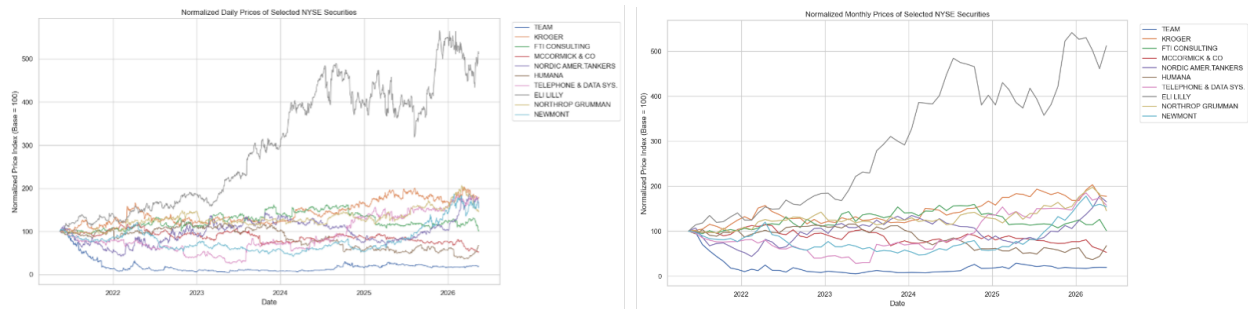


Figure 4: **Normalized price dynamics of selected NYSE securities using daily and monthly data.**

In Figure 4, the overall dispersion across securities is lighter than the Nasdaq case, showing what we said in previous questions about the more mature securities in portfolio.

Eli Lilly stands out as the dominant performer, exhibiting an extraordinary upward trajectory that reaches nearly five times its starting value by the end of the sample period, driven by the exceptional market enthusiasm for its drug pipeline. Northrop Grumman shows moderate positive trends, consistent with sustained defense spending and increasing of geopolitical tensions from 2022 onward. Kroger, McCormick & Co, FTI Consulting, and Humana display modest, relatively stable trajectories over the period, while Telephone & Data Systems and TEAM show persistent negative performance, falling well below their starting values. Nordic American Tankers exhibits sharp cyclical swings.

The visual inspection confirms that the correlation-based selection criterion worked as intended. Neither the Nasdaq nor the NYSE portfolio shows a uniform pattern of co-movement across all its members. This divergence across price trajectories is the visual expression of low average pairwise correlation, which is the core property that motivated the selection process for securities.

5 Q7&Q8 — Mean-Variance Optimal Portfolio Allocation: NYSE and Nasdaq Portfolios

The mean-variance framework developed by Markowitz provides the theoretical foundation for the portfolio construction exercises in this and the following sections. The idea is the following: given a set of assets with known expected returns and a covariance structure, there exists a portfolio that maximizes the reward per unit of risk taken. In practice, the weights of this portfolio are chosen by maximizing the Sharpe Ratio. Portfolio expected return, variance, and volatility are computed as:

$$E(R_p) = w^\top \mu$$

$$\sigma_p^2 = w^\top \Sigma w \quad \sigma_p = \sqrt{w^\top \Sigma w}$$

where w is the vector of portfolio weights, μ is the vector of expected returns, and Σ is the variance-covariance matrix. The Sharpe Ratio is:

$$SR = \frac{E(R_p) - R_f}{\sigma_p}$$

The annual risk-free rate is set at 2%, as is also made clear in Q18.

The full-investment constraint requires portfolio weights to sum to one, but no non-negativity constraint is imposed at this stage, which means that short selling is therefore allowed. The daily

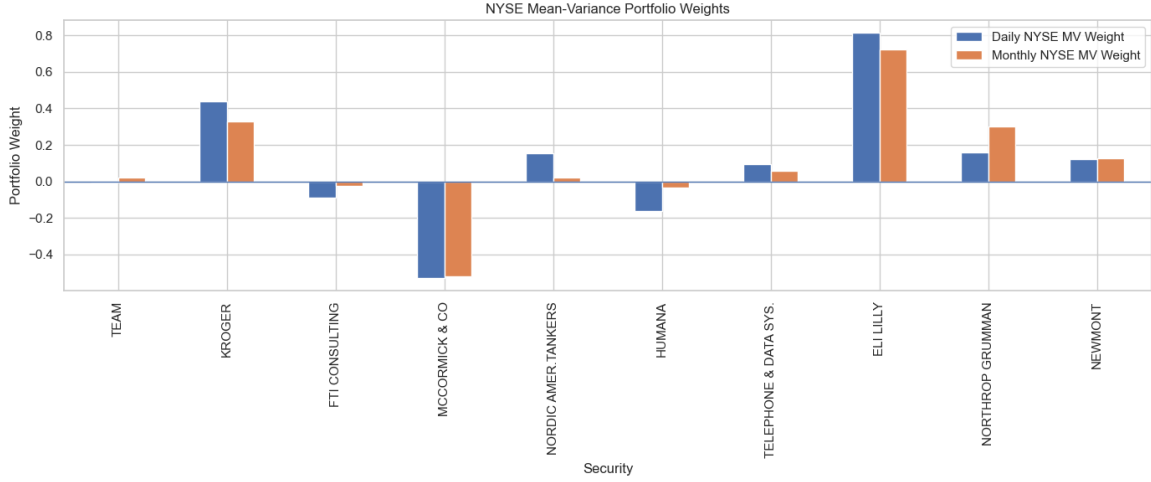


Figure 5: NYSE Mean-Variance Portfolio Weights

NYSE optimization assigns the largest positive weight to Eli Lilly (81.4%), followed by Kroger (43.8%), Northrop Grumman (15.8%), Nordic American Tankers (15.7%), Newmont (12.5%), and Telephone & Data Systems (9.4%). McCormick & Co, Humana, FTI Consulting, and TEAM receive negative weights, meaning the optimizer takes short positions in these securities to hedge exposures or improve the portfolio Sharpe Ratio. The short positions reflect the optimizer's assessment that these securities contribute negatively to the portfolio's risk-adjusted performance given the estimated covariance structure. The resulting daily NYSE portfolio delivers an expected return of approximately 0.197% per day, a volatility of 2.15%, and a Sharpe Ratio of 0.088.

The monthly NYSE optimization shifts the allocation substantially. Eli Lilly (72.3%) and Kroger (32.6%) remain the dominant long positions, now joined by Northrop Grumman (29.9%), Newmont (12.8%), and Telephone & Data Systems (5.7%). McCormick & Co, Humana, and FTI Consulting move to negative weights. This change is useful to understand the high sensitivity of mean-variance optimization to estimated input parameters, which change when returns are measured at different frequencies. The monthly NYSE portfolio achieves an expected return of

approximately 3.64% per month, a volatility of 7.61%, and a Sharpe Ratio of 0.457, substantially higher than the daily one.

The daily Nasdaq optimization concentrates heavily in Gilead Sciences (126.2%) and Exelon (86.2%), with positive allocations also to CME Group (66.5%), Quantum Computing (26.3%), PepsiCo (12.7%), Regeneron Pharmaceuticals (11.5%), and Oramed Pharmaceuticals (10.9%). Kraft Heinz, Astrotech, and ABVC Biopharma receive negative weights. The dominance of Gilead Sciences and Exelon is striking: at daily frequency, their return series exhibit a favorable combination of positive mean return and relatively contained variance, making them the optimizer’s preferred long positions by a wide margin. The extreme allocations highlight one of the most well-documented pathologies of unconstrained mean-variance optimization: the tendency to generate highly concentrated, unintuitive portfolios that are very sensitive to small changes in the input estimates. The daily Nasdaq portfolio delivers an expected return of

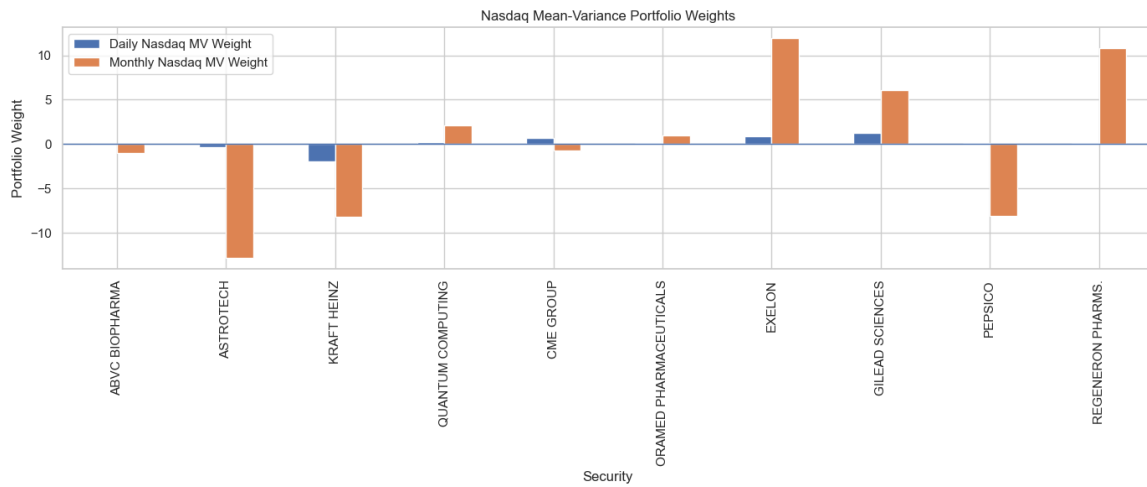


Figure 6: **Nasdaq Mean-Variance Portfolio Weights**

approximately 0.364% per day, a volatility of 4.19%, and a Sharpe Ratio of 0.085, slightly below the NYSE daily Sharpe Ratio despite the higher absolute return, because the Nasdaq universe carries meaningfully higher volatility.

The monthly Nasdaq optimization produces an even more extreme concentration, to the point of becoming economically implausible. Exelon receives a weight of approximately 1190%, followed by Regeneron Pharmaceuticals (1083%), Gilead Sciences (605%), and Quantum Computing (211%). A large group of securities, Astrotech, Kraft Heinz, PepsiCo, ABVC Biopharma, and CME Group, receive strongly negative weights, with Astrotech at -1285% and Kraft Heinz at -819% . The total long allocation far exceeds 100% and is offset by equally extreme short positions. The monthly Nasdaq portfolio achieves an expected return of approximately 97.5% per month, a volatility of 220.6%, and a Sharpe Ratio of 0.441. These extreme values are the direct consequence of the well-known instability of unconstrained mean-variance optimization when applied to a small sample of only 60 monthly observations with a high condition number of the covariance matrix (≈ 299). With so few data points and high estimation noise, particularly driven by the extreme volatility of securities like Quantum Computing (53% monthly std) and ABVC Biopharma (27%) , the optimizer exploits estimation error aggressively, producing

allocations that are mathematically valid but completely impractical. This result serves primarily as a theoretical benchmark and motivates the introduction of the non-negativity constraint in Questions 9 and 10.

The first thing to note is that both portfolios assign extreme weights to one or two securities, especially at monthly frequencies. This doesn't mean that these securities are particularly good investments but means that unconstrained mean-variance optimization applied to small samples with noise tends to identify whichever security has the most favorable recent history and allocate to it as aggressively as the covariance structure permits.

Then, it's important to notice that the Sharpe Ratio is higher at monthly frequency than at daily frequency for both portfolios. Part of this reflects the mathematical scaling of returns and volatility over time, but there's also a part that reflects the greater noise in daily return estimates.

At last, the NYSE presents a slightly higher Sharpe Ratio than the Nasdaq at daily frequency (0.088 vs 0.085), while both are comparable at monthly frequency. This suggests that the NYSE portfolio offers a marginally more favorable risk-adjusted environment at short horizons.

6 Q9&Q10 — Long-Only Mean-Variance Portfolios: Nasdaq and NYSE

The unconstrained portfolios of Q7&Q8 demonstrated both the power and the limits of the classical mean-variance framework. On one hand, the optimizer successfully identified combinations of assets that maximize the Sharpe Ratio given the estimated inputs. On the other hand, the resulting allocations were highly concentrated, relied heavily on short positions, and were driven to a significant degree by noise in the estimated expected returns.

The technique of short selling is not easy for an investor, implying borrowing costs, margin requirements, and regulatory constraints that make that portfolio very difficult to maintain in practice.

This section addresses these concerns by imposing a non-negativity constraint on all portfolio weights. The optimization problem becomes:

$$\max_w \quad SR = \frac{w^\top \mu - R_f}{\sqrt{w^\top \Sigma w}}$$

subject to:

$$\sum_{i=1}^N w_i = 1 \quad w_i \geq 0 \quad \forall i$$

The non-negativity constraint eliminates short positions and forces the optimizer to work only with long exposures. This restriction reduces the feasible set and therefore cannot improve upon the unconstrained Sharpe Ratio, the long-only portfolio will always achieve a Sharpe Ratio equal to or lower than its unconstrained counterpart.

The effect of the non-negativity constraint on the Nasdaq portfolio is immediate and substantial.

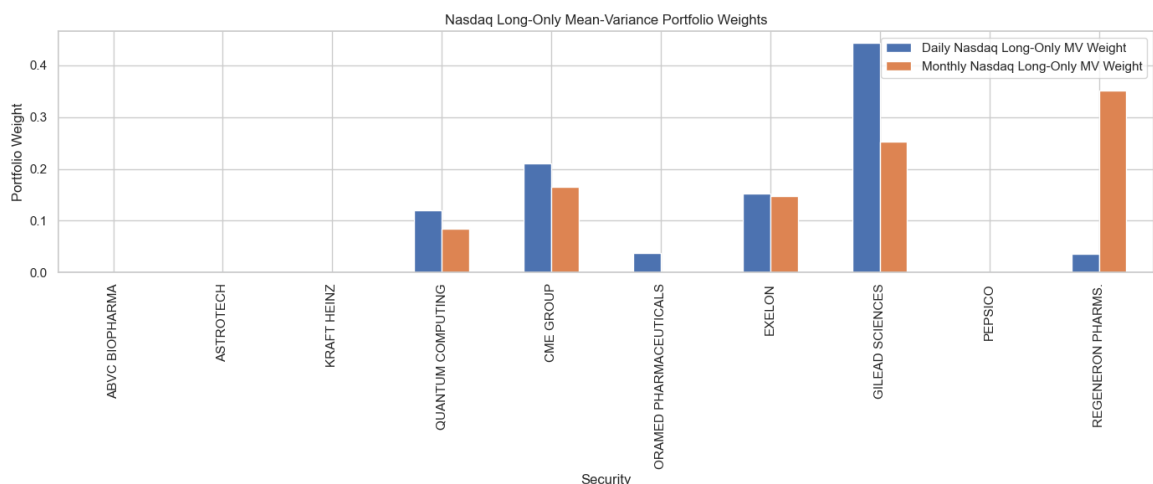


Figure 7: Nasdaq Long-Only Mean-Variance Portfolio Weights

The extreme allocations observed in Q7&Q8, weights of 1190% in Exelon and -1285% in Astrotech at monthly frequency, are no longer possible. Instead, the optimizer distributes weight across the securities that contributed positively in the unconstrained solution, while zeroing out those that previously received negative weights.

At daily frequency, the long-only allocation concentrates in Gilead Sciences (44.3%) and CME Group (21.1%), with smaller positive weights to Exelon (15.2%), Quantum Computing (12.1%), Oramed Pharmaceuticals (3.7%), and Regeneron Pharmaceuticals (3.5%). ABVC Biopharma, Astrotech, Kraft Heinz, and PepsiCo receive zero weight, they are excluded from the portfolio because without the ability to short them, the optimizer finds no reason to include them on the long side. At monthly frequency, the allocation shifts toward Regeneron Pharmaceuticals (35.2%), Gilead Sciences (25.3%), CME Group (16.5%), Exelon (14.7%), and Quantum Computing (8.4%). Again, ABVC Biopharma, Astrotech, Kraft Heinz, PepsiCo, and Oramed Pharmaceuticals receive zero weight. With daily frequencies, the Nasdaq long-only portfolio delivers a daily expected return of $\approx 0.089\%$ with volatility $\approx 1.39\%$ and Sharpe Ratio ≈ 0.058 , while with monthly frequencies it has an expected return of $\approx 1.594\%$ with volatility $\approx 5.27\%$ and Sharpe Ratio ≈ 0.271 .

The non-negativity constraint eliminates negative weights for the NYSE portfolio as well. At daily frequency, Eli Lilly (50.7%) and Kroger (21.4%) dominate the allocation, with smaller positive weights to Nordic American Tankers (10.5%), Newmont (7.5%), Telephone & Data Systems (5.5%), and Northrop Grumman (4.4%). TEAM, FTI Consulting, McCormick & Co, and Humana receive zero weight.

At monthly frequency, Eli Lilly (48.6%) remains dominant, followed by Kroger (17.6%), Northrop Grumman (19.3%), Nordic American Tankers (5.6%), Telephone & Data Systems (3.2%), Newmont (4.4%), TEAM (0.3%), and Humana (0.9%). FTI Consulting and McCormick & Co receive zero weight.

The NYSE long-only portfolio delivers a daily expected return of $\approx 0.107\%$ with volatility $\approx 1.26\%$ and Sharpe Ratio ≈ 0.079 , and a monthly expected return of $\approx 2.12\%$ with volatility $\approx 4.92\%$ and Sharpe Ratio ≈ 0.398 .

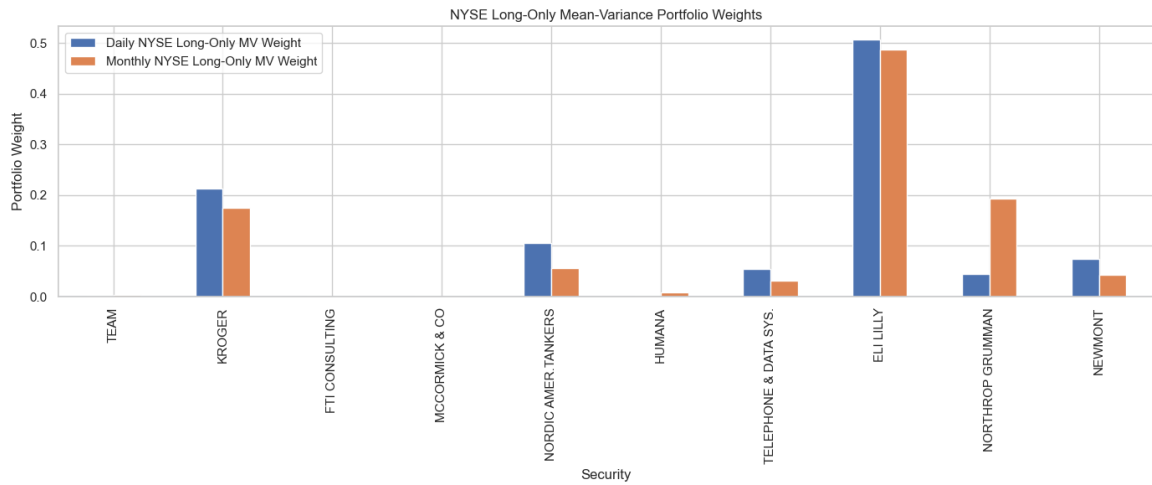


Figure 8: NYSE Long-Only Mean-Variance Portfolio Weights

Constraining weights enhances economic interpretability and aligns with practical institutional investment limitations. Both the Nasdaq and NYSE long-only portfolios remain relatively concentrated, with one or two securities dominating the allocation at both frequencies. This reflects the optimizer’s tendency to identify the securities with the most favorable risk-adjusted return and allocate to them as heavily as the long-only constraint permits.

7 Q11&Q12 — Descriptive Statistics of the Mean-Variance Portfolios

Portfolio	Mean	Std	Variance	Skewness	Exc. Kurtosis	Sharpe
Daily Unconstrained	0.197%	2.147%	0.000461	0.576	4.704	0.088
Monthly Unconstrained	3.645%	7.607%	0.005786	0.196	0.170	0.457
Daily Long-Only	0.107%	1.261%	0.000159	0.500	5.611	0.079
Monthly Long-Only	2.121%	4.915%	0.002416	0.212	0.429	0.398

Table 3: Descriptive Statistics — NYSE Mean-Variance Portfolios

Portfolio	Mean	Std	Variance	Skewness	Exc. Kurtosis	Sharpe
Daily Unconstrained	0.364%	4.194%	0.001759	0.091	7.023	0.085
Monthly Unconstrained	97.41%	220.47%	4.861	−0.029	0.637	0.441
Daily Long-Only	0.089%	1.394%	0.000194	1.017	5.541	0.058
Monthly Long-Only	1.594%	5.266%	0.002773	0.436	0.845	0.271

Table 4: Descriptive Statistics — Nasdaq Mean-Variance Portfolios

Unconstrained portfolios consistently achieve higher mean returns than their long-only counterparts. For NYSE, the daily unconstrained mean is nearly double the long-only daily mean, while at monthly frequency the gap is smaller. For Nasdaq, the daily unconstrained mean is roughly four times the long-only daily mean. The monthly unconstrained Nasdaq portfolio represents

an extreme case, which is a direct consequence of the numerical instability of the unconstrained optimizer, as discussed in Q7&Q8.

Regarding volatility, Nasdaq portfolios exhibit it systematically higher than NYSE portfolios at both frequencies. At daily ones, the Nasdaq unconstrained portfolio has almost twice the volatility of NYSE unconstrained. The long-only constraint reduces volatility substantially in both markets: from 4.194% to 1.394% for Nasdaq daily, and from 2.147% to 1.261% for NYSE daily. This confirms that the non-negativity constraint acts as an implicit risk-reduction mechanism by eliminating leveraged short-long combinations.

About the form of the distribution, the Nasdaq daily long-only portfolio shows the strongest positive skew, indicating a distribution with a longer right tail and potential for large positive returns. The NYSE portfolios display more moderate positive skewness across all specifications (0.196 to 0.576). Notably, the Nasdaq monthly unconstrained portfolio has near-zero skewness (-0.029), consistent with the near-symmetric extreme distribution produced by the unstable optimizer.

Finally, it can be seen that in both markets there's high excess kurtosis at daily frequency, with the Nasdaq daily unconstrained portfolio reaching 7.023 and the long-only portfolios of both markets around 5.5. This confirms the presence of heavy tails and a non-negligible probability of extreme daily returns. Monthly aggregation reduces kurtosis heavily, down to 0.170 for NYSE monthly unconstrained and 0.845 for Nasdaq monthly long-only, consistent with the central limit theorem smoothing out extreme events over longer horizons.

The NYSE long-only portfolios achieve higher Sharpe Ratios than their Nasdaq counterparts at both frequencies (0.079 vs 0.058 daily, 0.398 vs 0.271 monthly), confirming that the NYSE universe offers a more favorable risk-adjusted environment when short selling is excluded. Monthly portfolios consistently outperform daily ones in Sharpe Ratio terms for both markets, reflecting the aggregation of compounding returns and the reduction in daily noise.

Overall, these statistics confirm that volatility alone does not capture portfolio risk. Skewness and kurtosis are crucial for understanding tail risk and downside exposure. Long-only constraints produce realistic, implementable portfolios but with moderated expected returns, while unconstrained optimization leverages more extreme risk-return profiles, especially for speculative Nasdaq securities.

8 Q13&Q14 — Efficient Frontier: Nasdaq and NYSE Portfolios

The efficient frontier is the graphical representation of the set of portfolios that minimize volatility for a given level of expected return. Here, we show the long-only efficient frontiers for the selected Nasdaq and NYSE portfolios at daily and monthly frequencies. Portfolio expected return and variance are calculated as:

$$E(R_p) = w^\top \mu \quad \sigma_p^2 = w^\top \Sigma w$$

where w is the vector of portfolio weights, μ is the vector of expected asset returns, and Σ is the variance-covariance matrix. The Global Minimum Variance (GMV) portfolio represents the lowest-volatility feasible allocation, while the long-only Mean-Variance portfolio maximizes risk-adjusted return along the frontier.

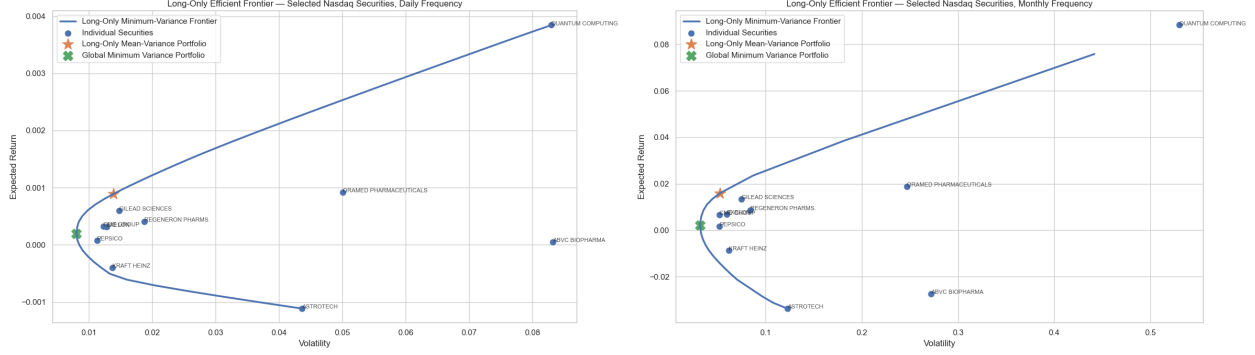


Figure 9: Daily and monthly Long-Only efficient frontier for Nasdaq.

Metric	Daily GMV	Monthly GMV
Expected Return	0.020%	0.206%
Volatility	0.807%	3.197%
Variance	0.000065	0.001022
Sharpe Ratio	0.015	0.012

Table 5: Global Minimum Variance Portfolio Performance for Nasdaq

The daily Nasdaq frontier spans a return range from 0.020% to 0.364% with volatility ranging from 0.807% to 8.301%. The GMV portfolio sits at the leftmost point of the frontier, representing the minimum-risk allocation achievable within the long-only constraint. Its Sharpe Ratio of 0.015 is notably low considering the one achieved in long-only mean-variance analysis before (four times higher). This is important to highlight because minimum-variance portfolio is not the same as the most efficient one, and an investor who simply minimizes variance leaves significant risk-adjusted return on the table. So, accepting a moderate increase in volatility relative to the GMV produces a substantially better risk-adjusted outcome, and the minimum-variance point is far from the most efficient allocation for an investor who cares about return per unit of risk.

Moving rightward along the frontier, the allocation progressively tilts toward higher-volatility speculative securities, Quantum Computing, Astrotech, and ABVC Biopharma, which dominate the upper end of the curve and are responsible for the steep increase in volatility beyond the midpoint.

The monthly Nasdaq frontier expands dramatically in scale, with the GMV sitting at return 0.206% and volatility 3.197%. The Sharpe Ratio of the monthly GMV is even lower than the daily one, consistent with the noisier return estimates at monthly frequency with only 60 observations. The upper end of the monthly frontier is dominated by the extreme volatility of speculative securities, pushing the right tail of the curve well beyond what is practically investable.

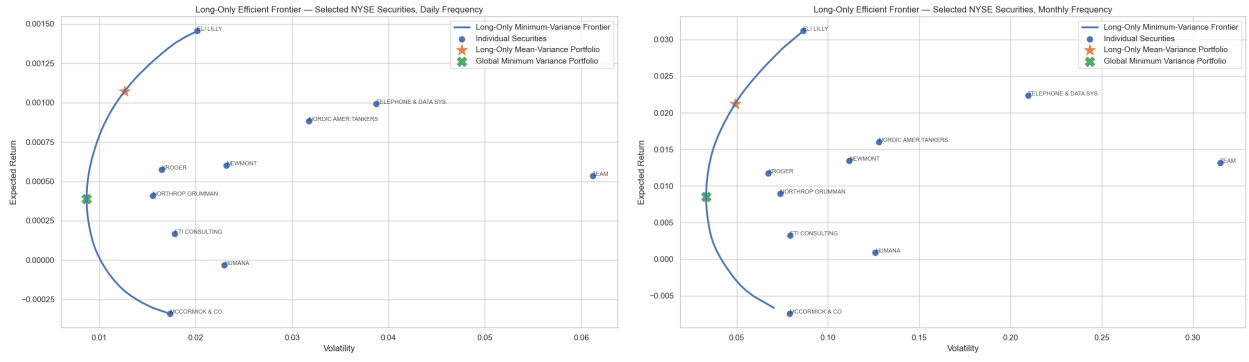


Figure 10: Daily and monthly Long-Only efficient frontier for NYSE.

Metric	Daily GMV	Monthly GMV
Expected Return	0.039%	0.856%
Volatility	0.869%	3.331%
Variance	0.000076	0.001109
Sharpe Ratio	0.036	0.207

Table 6: Global Minimum Variance Portfolio Performance for NYSE

The NYSE frontier shows a different situation; at daily frequency, the frontier shows a return that goes from 0.039% to 0.123%, with volatility ranging from 0.869% to 1.699%, both narrower and more compact than the Nasdaq counterpart. The GMV portfolio already achieves a Sharpe Ratio of 0.036, more than double the Nasdaq daily GMV, reflecting the more homogeneous and stable nature of the selected NYSE securities. The curve evolves gradually from the GMV to the long-only MV portfolio (return 0.107%, volatility 1.261%, Sharpe 0.079) without dramatic jumps — there are no extreme individual volatilities pulling the upper end of the frontier far to the right.

The monthly NYSE frontier spans return from 0.856% to 3.116% with volatility ranging from 3.331% and 8.644%. The GMV portfolio achieves a Sharpe Ratio of 0.207 which is by far the highest GMV Sharpe across all four frontier specifications confirming that the NYSE monthly universe offers the most favorable minimum-risk allocation in risk-adjusted terms. What is interesting to notice is that the monthly frontier retains a conservative structure compared with the Nasdaq, without showing an expansion into implausible territory, and the shape is more balanced thanks to a more balanced sectoral contribution.

Putting the two frontiers side by side, we note at first glance how wider and steeper the Nasdaq frontier is with respect to NYSE one. the contrast is clear. Furthermore, the Nasdaq frontier is more sensitive to the presence of speculative securities, a small number of high-volatility names that pull the upper end of the curve far to the right.

Across both markets, the GMV consistently achieves a lower Sharpe Ratio than the long-only MV portfolio, confirming that minimizing variance and maximizing risk-adjusted return are distinct objectives that lead to meaningfully different allocations.

Q15 — Descriptive Statistics of the US Market Indexes

This section compares the optimized NYSE and Nasdaq Mean-Variance portfolios with broad US market indexes, including both the Price and Total Return versions of the S&P 500 Composite index and the Nasdaq Composite index, using daily and monthly frequencies. For each index, returns are calculated as:

$$R_t = \frac{P_t}{P_{t-1}} - 1$$

The following descriptive statistics are computed for each index: mean return, standard deviation, variance, skewness, excess kurtosis, and Sharpe Ratio, allowing a direct comparison of the risk-return profiles of the optimized portfolios versus the passive market benchmarks.

	Mean	Std	Variance	Skewness	Exc. Kurt.	Sharpe
S&P 500 Price Index	0.000494	0.010441	0.000109	0.174626	7.605897	0.039730
S&P 500 Tot. Return	0.000550	0.010441	0.000109	0.176602	7.594266	0.045050
Nasdaq Composite Price	0.000516	0.013984	0.000196	0.197044	5.707995	0.031209
Nasdaq Composite Tot. Ret.	0.000544	0.013985	0.000196	0.197592	5.702706	0.033234
NYSE MV Portfolio	0.001974	0.021465	0.000461	0.576472	4.726987	0.088278
Nasdaq MV Portfolio	0.003640	0.041936	0.001759	0.090899	7.054383	0.084915

Table 7: Descriptive Statistics — Daily Frequency

	Mean	Std	Variance	Skewness	Exc. Kurt.	Sharpe
S&P 500 Price Index	0.010564	0.044109	0.001946	-0.223370	0.443638	0.201713
S&P 500 Tot. Return	0.011782	0.044108	0.001946	-0.216371	0.441497	0.229339
Nasdaq Composite Price	0.010636	0.058568	0.003430	-0.299928	0.788532	0.153142
Nasdaq Composite Tot. Ret.	0.011260	0.058632	0.003438	-0.294545	0.776496	0.163619
NYSE MV Portfolio	0.036446	0.076067	0.005786	0.201011	0.291687	0.457224
Nasdaq MV Portfolio	0.974528	2.205587	4.864613	-0.029895	0.800288	0.441090

Table 8: Descriptive Statistics — Monthly Frequency

Both optimized portfolios deliver substantially higher mean returns than the passive benchmarks at daily frequency. The NYSE MV portfolio outperforms the S&P 500 Total Return by a factor of nearly four, while the Nasdaq MV portfolio outperforms the Nasdaq Composite Total Return by a factor of nearly seven. At monthly frequency, it happens the same thing for the NYSE against S&P 500, while the Nasdaq monthly MV portfolio becomes a useless comparison due to its numerical instability, as discussed in Q7&Q8.

The optimized portfolios carry higher volatility than the broad market indexes at daily frequency, which is expected given their more concentrated allocations. Intuitively, we can say that the higher volatility of the optimized portfolios is the price paid for their higher mean returns.

However, the most important data to analyze is the Sharpe ratio. From the tables, it's possible to see by how much. At daily frequency, the NYSE MV portfolio (0.088) and the Nasdaq MV portfolio (0.085) both outperform all four market indexes, whose Sharpe Ratios range from

0.031 to 0.045 at daily frequencies. Shifting the view on monthly data, the point doesn't change: the NYSE MV portfolio (0.457) substantially outperforms the S&P 500 Total Return (0.229) and the Nasdaq Composite Total Return (0.164). This confirms that the mean-variance optimization has successfully identified allocations that improve upon the passive market benchmarks on a risk-adjusted basis.

The market indexes exhibit mild positive skewness at daily frequency and mild negative at monthly, while shows exhibit very high excess kurtosis at daily frequency at this pattern is smoothed out at monthly aggregation. Regarding the portfolios, the NYSE MV shows more pronounced positive skewness at both frequencies (0.576 daily, 0.201 monthly), indicating a distribution with a longer right tail relative to the benchmarks.

9 Q16&Q17 — Beta Estimation: NYSE and Nasdaq Securities and Portfolios

This section estimates beta for each selected NYSE and Nasdaq security and for the corresponding optimized long-only Mean-Variance portfolios. Beta measures the sensitivity of a security or portfolio to movements in the relevant market benchmark:

$$\beta_i = \frac{\text{Cov}(R_i, R_m)}{\text{Var}(R_m)}$$

where β_i is the beta of security or portfolio i , R_i is its return, and R_m is the return of the market benchmark. The S&P 500 Total Return Index is used as the benchmark for NYSE securities, and the Nasdaq Composite Total Return Index is used for Nasdaq securities. Beta is estimated using the sample covariance with $ddof = 1$ to ensure unbiasedness.

Security	Daily Beta	Monthly Beta
TEAM	0.9798	1.6132
Kroger	0.1172	0.0715
FTI Consulting	0.3020	−0.1050
McCormick & Co	0.2028	0.6822
Nordic Amer. Tankers	0.5726	0.3503
Humana	0.4238	0.1643
Telephone & Data Sys.	0.6631	0.2559
Eli Lilly	0.6064	0.1857
Northrop Grumman	0.2256	−0.3251
Newmont	0.5436	0.1781
Portfolio	0.5930	−0.2174

Table 9: **Beta Estimates — NYSE Securities and Portfolio (vs S&P 500 Total Return)**

The NYSE beta estimates reveal a predominantly defensive portfolio universe. At daily frequency, all ten securities have positive betas, ranging from 0.117 of Kroger to 0.978 of TEAM. Almost all the securities fall well below 1, displaying low sensitivity to the S&P 500, consistent with their consumer staples, defense, and professional services profiles. TEAM is the only

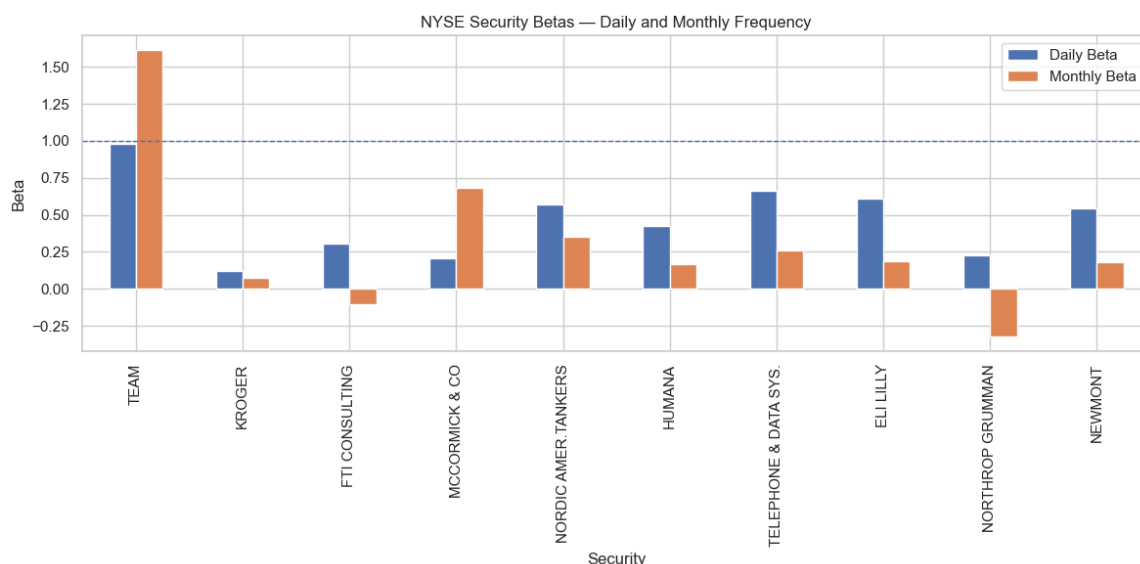


Figure 11: NYSE Security Betas — Daily and Monthly Frequencies

security approaching a beta of 1 at daily frequency, reflecting its higher sensitivity to broad market movements.

Things change drastically talking about monthly frequency. TEAM's beta rises to 1.613, consistent with the amplification of its technology growth exposure over longer horizons. Some securities like Northrop Grumman and FTI Consulting turns negative, reflecting the counter-cyclical nature of defense spending and restructuring advisory demand over longer time horizons. These negative monthly betas reflect the fact that government defense budgets and corporate distress activity tend to increase precisely when the broader market is under stress.

The NYSE portfolio daily beta of 0.593 indicates moderate market exposure, because the portfolio moves roughly half as much as the S&P 500 on any given day. The concentration of the unconstrained daily portfolio in Eli Lilly (81%) drives this result, as Eli Lilly's individual beta of 0.606 dominates the weighted average. At monthly frequency, the portfolio beta turns negative (-0.217), driven by the large negative weight on McCormick & Co combined with Northrop Grumman's negative monthly beta.

This is the main tool for showing that the unconstrained monthly NYSE portfolio, through its short positions, effectively takes on a counter-market exposure over longer horizons.

Security	Daily Beta	Monthly Beta
ABVC Biopharma	−0.1556	−1.2028
Astrotech	−0.0390	0.1226
Kraft Heinz	0.0072	−0.0346
Quantum Computing	−0.0144	0.0588
CME Group	−0.0132	0.0932
Oramed Pharmaceuticals	0.0499	1.1965
Exelon	−0.0124	0.2057
Gilead Sciences	−0.0037	0.1104
PepsiCo	0.0097	−0.1266
Regeneron Pharmaceuticals	−0.0116	−0.1550
Portfolio	−0.0189	3.6393

Table 10: Beta Estimates — Nasdaq Securities and Portfolio (vs Nasdaq Composite Total Return)

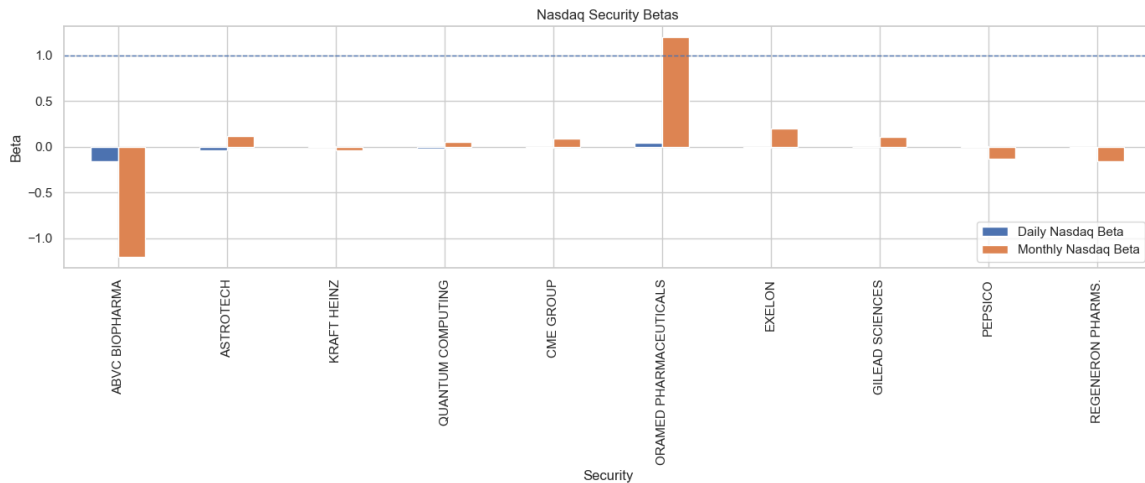


Figure 12: Nasdaq Security Betas— Daily and Monthly Frequencies

The Nasdaq beta estimates are striking because, at daily frequency, almost all selected securities have betas close to zero or negative relative to the Nasdaq Composite Total Return Index. The resulting portfolio daily beta of -0.019 is essentially zero.

This outcome is the direct consequence of the selection methodology. The securities were chosen precisely because they exhibit the lowest average absolute correlation with the Nasdaq universe. A low correlation with the Nasdaq market naturally translates into a beta close to zero or negative relative to the Nasdaq Composite, which was the driver of our portfolio selection, instead trying to build up one that tracks the Nasdaq market.

At monthly frequency, the betas become more dispersed. ABVC Biopharma reaches -1.203 and Oramed Pharmaceuticals reaches 1.197 , reflecting the high estimation noise inherent in monthly beta estimation with only 60 observations. The huge portfolio monthly beta showed in Table 10 is driven by the extreme unconstrained weights of the monthly optimizer as shown in Q8, because the monthly unconstrained Nasdaq portfolio assigns weights of over 1000% to Exelon and Regeneron. Given this, despite their low individual betas produce a large aggregate systematic exposure when leveraged to such extremes.

This is yet another piece of evidence that has emerged as the paper has progressed that the monthly unconstrained Nasdaq portfolio is numerically unstable and should not be interpreted as a practically meaningful allocation.

10 Q18&Q19 — Security Market Line Verification: NYSE and Nasdaq Portfolios

This section verifies the Security Market Line (SML) for two selected securities from each portfolio and for the corresponding long-only Mean-Variance portfolios. The analysis is based on the Capital Asset Pricing Model (CAPM), which links expected return to systematic risk:

$$E(R_i) = R_f + \beta_i [E(R_m) - R_f]$$

where R_f is the risk-free rate (2% annually, adjusted to daily and monthly equivalents), β_i is the beta of the security or portfolio, and $E(R_m)$ is the expected return of the market benchmark. The SML is verified by comparing the actual mean return of each security with its CAPM-implied return. A positive difference, Actual minus CAPM, indicates that the security plots above the SML, it has delivered more return than CAPM would predict given its systematic risk, generating a positive alpha. A negative difference indicates the opposite.

The two selected NYSE securities are Newmont and TEAM. We chose these two because they represent quite different behavior of the beta within the NYSE selection. Newmont has a daily beta of 0.544, while TEAM has a daily beta of 0.980, the closest to 1 among all selected NYSE securities. By doing this, we allow a useful verification of the SML across different systematic risk levels.

For Nasdaq, we chose ABVC Biopharma and Oramed Pharmaceutical. Obviously, the criteria was the same of NYSE selection, choosing contrasting values of beta. ABVC Biopharma has the most negative daily beta in the portfolio (-0.156), making it the security most inversely related to the Nasdaq Composite, while Oramed Pharmaceuticals has the highest positive daily beta among the selected securities.

Security	Beta	Actual Return	CAPM Return	Difference
Newmont	0.5436	0.000602	0.000335	+0.000267
TEAM	0.9798	0.000536	0.000540	−0.000004
NYSE Long-Only Portfolio	0.4799	0.001071	0.000305	+0.000766

Table 11: Daily NYSE SML Verification (vs S&P 500 Total Return)

Security	Beta	Actual Return	CAPM Return	Difference
Newmont	0.1781	0.013448	0.003469	+0.009979
TEAM	1.6132	0.013150	0.017985	−0.004835
NYSE Long-Only Portfolio	0.0828	0.021214	0.002504	+0.018710

Table 12: Monthly NYSE SML Verification (vs S&P 500 Total Return)

At daily frequency, Newmont plots above the SML with a slightly positive alpha, confirming that over the sample period it has delivered more return than CAPM would predict given its beta of 0.544. This is consistent with period analyzed, since during periods of market stress Newmont tends to outperform its CAPM-implied return because investors bid up gold-related assets beyond what systematic risk alone would justify.

TEAM is almost exactly on the SML at daily frequency, and this is really interesting to see because it is an almost perfect verification of the CAPM for this security: having a beta close to 1, TEAM’s actual daily return closely matches what the market risk premium would predict.

The NYSE long-only portfolio plots well above the SML at daily frequency, indicating that the portfolio has generated a meaningful positive alpha relative to the S&P 500. This happens because we chose securities that move independently of the market, so the portfolio has captured idiosyncratic returns that are not explained by systematic risk.

At monthly frequency, all these patterns are more pronounced. Newmont’s positive alpha is substantially larger, reflecting the stronger gold price appreciation over longer horizons. TEAM, by contrast, plots below the SML at monthly frequency, its actual return falls short of what its elevated monthly beta of 1.613 would imply under CAPM. This is consistent with TEAM’s poor cumulative performance documented in Q6, where it persistently underperformed relative to its starting value. The portfolio again plots far above the SML confirming the persistence of the positive alpha across both frequencies.

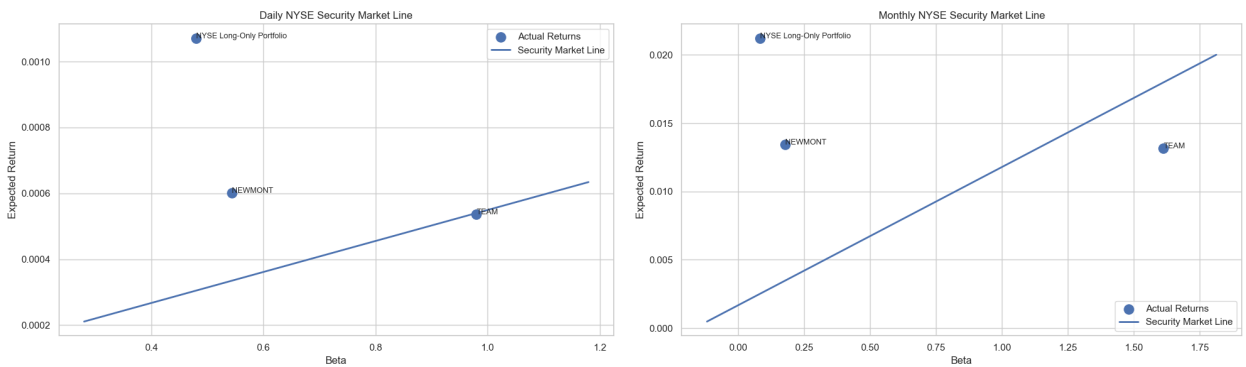


Figure 13: Daily and monthly NYSE Security Market Line.

Security	Beta	Actual Return	CAPM Return	Difference
ABVC Biopharma	−0.1556	0.000047	0.000007	+0.000040
Oramed Pharmaceuticals	0.0499	0.000921	0.000103	+0.000818
Nasdaq Long-Only Portfolio	−0.0066	0.000893	0.000076	+0.000817

Table 13: Daily Nasdaq SML Verification (vs Nasdaq Composite Total Return)

Security	Beta	Actual Return	CAPM Return	Difference
ABVC Biopharma	-1.2028	-0.027490	-0.009872	-0.017618
Oramed Pharmaceuticals	1.1965	0.018695	0.013146	+0.005549
Nasdaq Long-Only Portfolio	0.0239	0.015938	0.001896	+0.014041

Table 14: Monthly Nasdaq SML Verification (vs Nasdaq Composite Total Return)

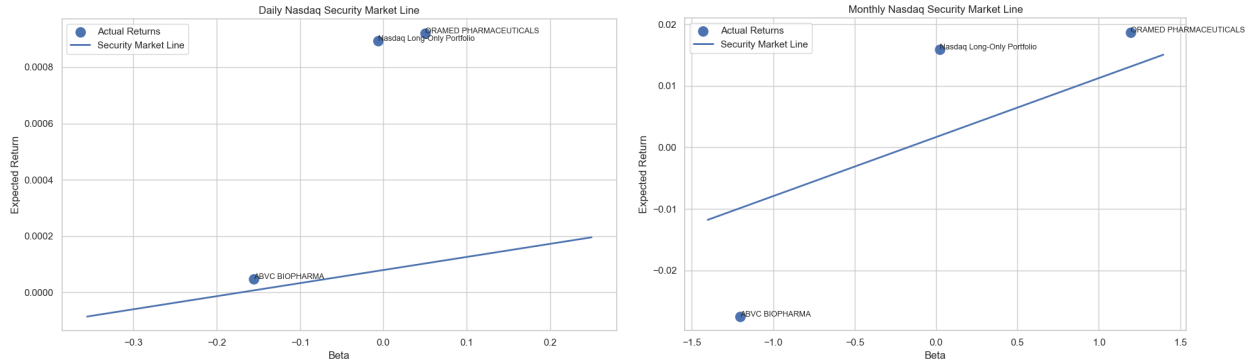


Figure 14: **Daily and monthly Nasdaq Security Market Line.**

Here instead, both selected securities and the portfolio plot above the SML at daily frequency. ABVC Biopharma has a near-zero actual return but its negative beta means that the CAPM predicts an even lower return, the positive difference is small in absolute terms but reflects the fact that this security has marginally outperformed its CAPM prediction. Oramed Pharmaceuticals plots well above the SML, having substantially more return than its near-zero beta would predict under CAPM. The portfolio itself also plots above the SML, consistent with the positive alpha observed for the NYSE portfolio.

Different things happen when we go on monthly frequency. ABVC Biopharma plots below the SML, consistent with the poor long-term performance of ABVC Biopharma documented in Q6. Oramed Pharmaceuticals and the portfolio remain above the SML, driven by the strong positive contributions of Gilead Sciences and CME Group in the long-only allocation.

Definitively, in all situations of markets and frequencies, the long-only portfolios consistently plot above the SML, indicating positive alpha generation relative to the respective market benchmarks.

This is the logic consequence of the selection methodology we adopted, based on a diversification goal, pursued through the choice of securities with low average absolute correlation with the market.

As a result, we end up with portfolios that can capture idiosyncratic return premiums that are not explained by systematic risk.

The individual securities, by contrast, show more mixed results: some plot above the SML (Newmont, Oramed), some are approximately on it (TEAM daily), and some plot below it (TEAM monthly, ABVC monthly), reflecting the inherent uncertainty of CAPM predictions for individual securities over finite sample periods.

11 Q20&Q21 — Black-Litterman Allocation: NYSE and Nasdaq Portfolios

The Black-Litterman model addresses one of the most known limitations of classical mean-variance optimization: its extreme sensitivity to the estimated expected return vector. Rather than using historical sample means directly as inputs, the Black-Litterman approach combines a prior equilibrium return vector with investor views through a Bayesian updating procedure, producing a posterior expected return vector that is more stable and economically interpretable.

The prior equilibrium return vector Π is derived from the assumption that the market is in equilibrium, using an equal-weight benchmark allocation in place of market-capitalization weights, which are not available in the dataset:

$$\Pi = \lambda \Sigma w_0$$

where $\lambda = 2.5$ is the risk aversion parameter and w_0 is the equal-weight benchmark vector. The posterior expected return vector is then computed as:

$$\mu_{BL} = [(\tau \Sigma)^{-1} + P^\top \Omega^{-1} P]^{-1} [(\tau \Sigma)^{-1} \Pi + P^\top \Omega^{-1} Q]$$

and the posterior covariance matrix as:

$$\Sigma_{BL} = \Sigma + [(\tau \Sigma)^{-1} + P^\top \Omega^{-1} P]^{-1}$$

where $\tau = 0.05$ controls the uncertainty around the prior, P is the view matrix, Q is the vector of view returns, and $\Omega = \text{diag}(P \tau \Sigma P^\top)$ is the view uncertainty matrix. The parameter $\tau = 0.05$ is a standard choice in the literature, reflecting the assumption that the prior equilibrium is relatively well-established and the views carry meaningful but limited weight relative to it.

We have specified four views for the NYSE portfolio, using only assets present in the selected universe:

- **Absolute View 1:** TEAM is expected to generate a daily return of 0.09% (monthly: 1.80%). This view reflects a positive outlook on growth trajectory.
- **Absolute View 2:** Kroger is expected to generate a daily return of 0.04% (monthly: 0.80%), consistent with its stable consumer staples profile and resilient demand.
- **Relative View 1:** TEAM is expected to outperform FTI Consulting by 0.05% daily (1.00% monthly), reflecting a view that technology-driven growth will recover faster than professional services.
- **Relative View 2:** McCormick & Co is expected to outperform Nordic American Tankers by 0.02% daily (0.40% monthly), reflecting a view that consumer staples offer more stable returns than volatile shipping stocks.

Security	Prior Π	Posterior μ_{BL}
TEAM	0.001052	0.000881
Kroger	0.000153	0.000280
FTI Consulting	0.000167	0.000190
McCormick & Co	0.000119	0.000194
Nordic Amer. Tankers	0.000363	0.000206
Humana	0.000199	0.000210
Telephone & Data Sys.	0.000531	0.000546
Eli Lilly	0.000195	0.000204
Northrop Grumman	0.000153	0.000172
Newmont	0.000232	0.000239

Table 15: **NYSE Prior vs Posterior Expected Returns — Daily**

Security	Prior Π	Posterior μ_{BL}
TEAM	0.034230	0.021248
Kroger	0.001368	0.004810
FTI Consulting	0.002382	0.002612
McCormick & Co	0.003315	0.005029
Nordic Amer. Tankers	0.008932	0.004875
Humana	0.003545	0.001468
Telephone & Data Sys.	0.016703	0.014882
Eli Lilly	0.002597	0.003276
Northrop Grumman	0.002828	0.002869
Newmont	0.006888	0.007151

Table 16: **NYSE Prior vs Posterior Expected Returns — Monthly**

The prior vs posterior tables reveal how the views shift expected returns relative to the equilibrium. The most notable movements at daily frequency are Kroger and McCormick & Co, which are driven by the absolute view on Kroger and the relative view favoring McCormick over Nordic American Tankers. Conversely, Nordic American Tankers is pulled down drastically by the relative view placing it below McCormick. TEAM's posterior is lower than the prior despite the positive absolute view and this reflects the Bayesian shrinkage toward the prior, which was already high for TEAM due to its elevated volatility.

Security	Daily BL Weight	Monthly BL Weight
TEAM	0.1033	0.0786
Kroger	0.2849	0.3403
FTI Consulting	0.0686	0.0395
McCormick & Co	0.1192	0.1291
Nordic Amer. Tankers	0.0248	0.0261
Humana	0.0813	0.0236
Telephone & Data Sys.	0.1239	0.1481
Eli Lilly	0.0657	0.0421
Northrop Grumman	0.0438	0.0000
Newmont	0.0844	0.1727

Table 17: **NYSE Black-Litterman Portfolio Weights**

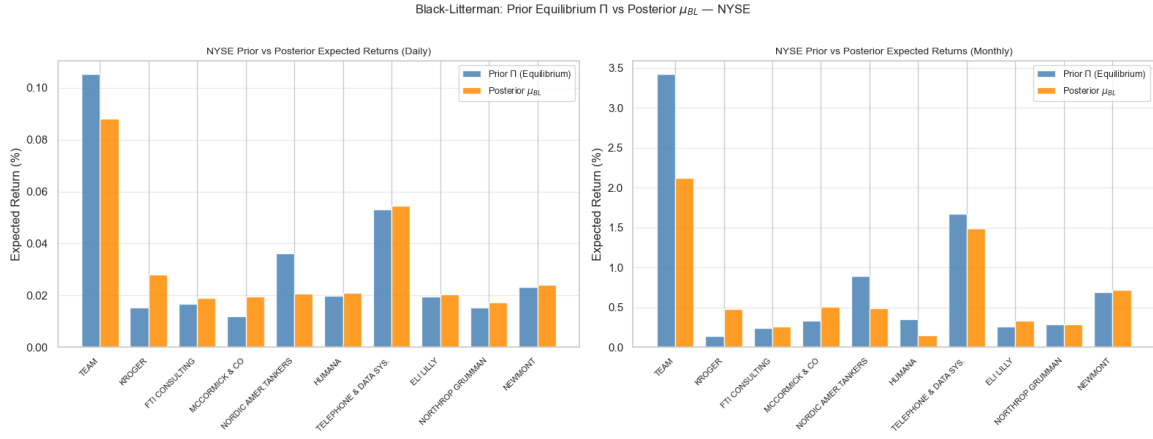


Figure 15: **Prior equilibrium vs. Posterior — NYSE**

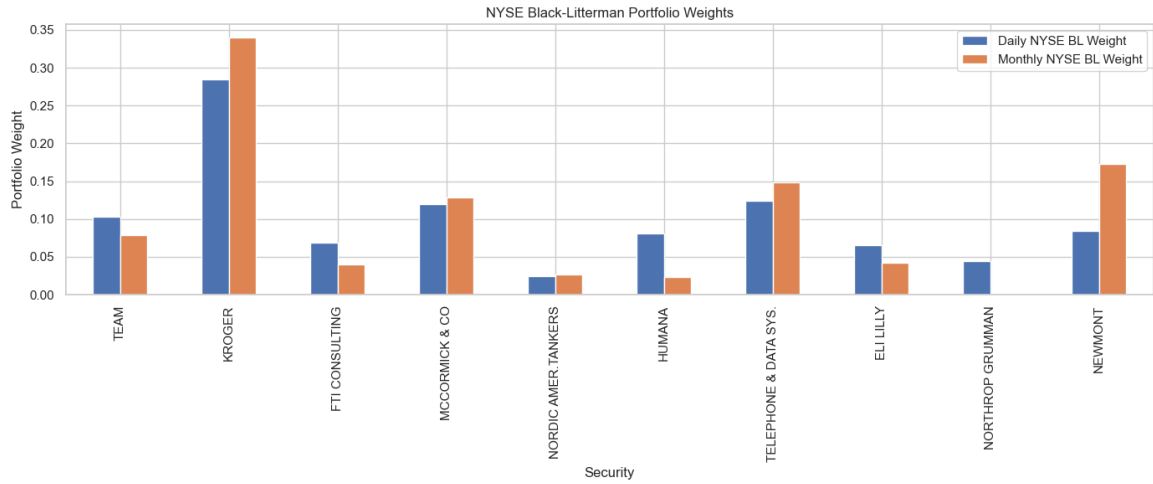


Figure 16: **NYSE Black-Litterman Portfolio Wreights**

The resulting BL portfolio is, as expected, far more diversified than the long-only MV portfolio of Q9. First thing to notice is that now all the ten securities receive positive weight at daily frequency, compared with only six in the MV long-only case. It's interesting to see what happened to Eli Lilly security, which dominated the MV long-only portfolio at 50.7%, receives only 6.6% in the BL portfolio: reason is that posterior expected returns are more evenly distributed across assets, so the optimizer spreads weight more broadly.

Metric	Daily BL	Monthly BL
Mean	0.0497%	1.159%
Std	1.170%	6.071%
Variance	0.000137	0.003685
Skewness	0.802	0.748
Excess Kurtosis	8.163	1.297
Sharpe Ratio	0.036	0.163

Table 18: **NYSE Black-Litterman Portfolio Statistics**

Comparing with the long-only MV portfolio, we can note that BL daily Sharpe Ratio is lower than the MV long-only (0.079), but the portfolio is substantially more diversified and less

sensitive to estimation noise. Also at monthly frequency happens the same thing, because the BL Sharpe is still lower than the MV long-only (0.398) but it gains in diversification.

We specified four views also for the Nasdaq portfolio:

- **Absolute View 1:** ABVC Biopharma is expected to generate a daily return of 0.10% (monthly: 2.00%), reflecting a positive view on a potential clinical catalyst.
- **Absolute View 2:** Astrotech is expected to generate a daily return of 0.03% (monthly: 0.60%), reflecting a modest positive outlook on its early-stage technology developments.
- **Relative View 1:** ABVC Biopharma is expected to outperform Kraft Heinz by 0.06% daily (1.20% monthly), reflecting a view that biotechnology upside exceeds consumer staples returns.
- **Relative View 2:** Quantum Computing is expected to outperform CME Group by 0.04% daily (0.80% monthly), reflecting a speculative view on quantum technology momentum.

Security	Prior Π	Posterior μ_{BL}
ABVC Biopharma	0.001899	0.001203
Astrotech	0.000634	0.000450
Kraft Heinz	0.000129	0.000140
Quantum Computing	0.002110	0.001286
CME Group	0.000110	0.000118
Oramed Pharmaceuticals	0.000920	0.000835
Exelon	0.000106	0.000114
Gilead Sciences	0.000169	0.000163
PepsiCo	0.000111	0.000119
Regeneron Pharmaceuticals	0.000212	0.000201

Table 19: **Nasdaq Prior vs Posterior Expected Returns — Daily**

Security	Prior Π	Posterior μ_{BL}
ABVC Biopharma	0.019094	0.017646
Astrotech	0.007138	0.006190
Kraft Heinz	0.000778	0.001919
Quantum Computing	0.067047	0.038145
CME Group	0.001458	0.001674
Oramed Pharmaceuticals	0.021993	0.020190
Exelon	0.002098	0.002555
Gilead Sciences	0.003426	0.003490
PepsiCo	0.000993	0.001890
Regeneron Pharmaceuticals	-0.001021	0.000982

Table 20: **Nasdaq Prior vs Posterior Expected Returns — Monthly**

Unlike what we saw with the NYSE, the Nasdaq is undergoing a systematic downward revision of expected returns relative to the prior. The most notable reductions are for ABVC Biopharma and Quantum Computing.

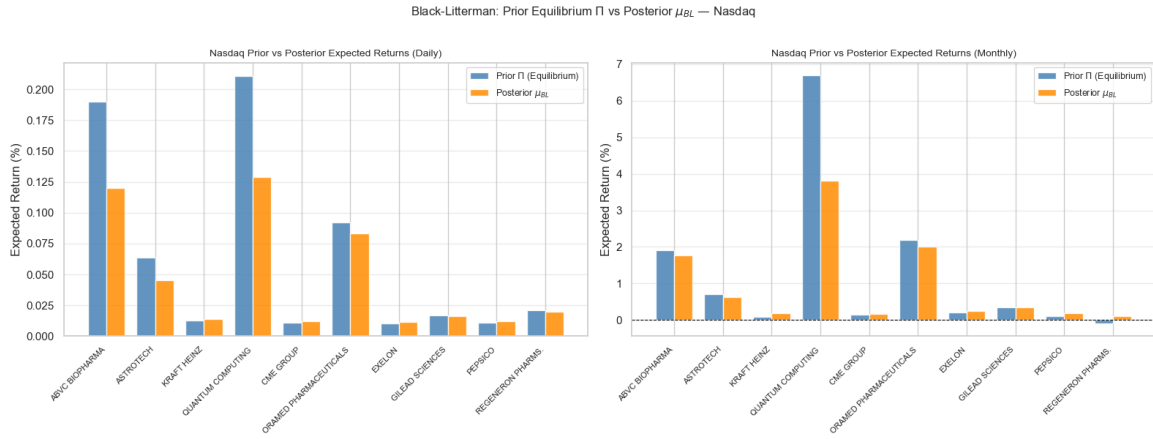


Figure 17: **Prior equilibrium vs. Posterior** — Nasdaq

This is quite a counter-intuitive result, because the views on these assets were positive but yet their posterior returns fell, is once again determined by the typical Bayesian mechanism: given that the prior equilibrium returns were already very high due to their elevated volatility the views, although positive, are not high enough to support them. At the end, posterior balances the two and produces a moderate downward revision.

Security	Daily BL Weight	Monthly BL Weight
ABVC Biopharma	0.1280	0.2236
Astrotech	0.1202	0.1236
Kraft Heinz	0.1121	0.0000
Quantum Computing	0.1204	0.1041
CME Group	0.0100	0.0000
Oramed Pharmaceuticals	0.2035	0.2601
Exelon	0.0363	0.0720
Gilead Sciences	0.0908	0.2127
PepsiCo	0.0391	0.0000
Regeneron Pharmaceuticals	0.1398	0.0038

Table 21: Nasdaq Black-Litterman Portfolio Weights

The same thing is happening here as with the NYSE portfolio earlier: it is more diversified than the long-only MY portfolio, and here too all ten securities are assigned a positive weighting at daily frequencies, with Oramed Pharmaceuticals and Regeneron. It can therefore be seen that the situation is quite different from the case mentioned, where the main weights were concentrated in Gilead Sciences and CME Group, which together accounted for around 65%. At monthly frequency, the allocation shifts toward ABVC Biopharma, Oramed Pharmaceuticals, and Gilead Sciences. In this case, however, it should be noted that, when looking at monthly data, it is more common to find securities with a zero weighting. In fact, whereas in the case of the NYSE it was only Northrop Grumman that did not receive one, here there are as many as three: CME Group, Kraft Heinz, and PepsiCo receiving zero weight.

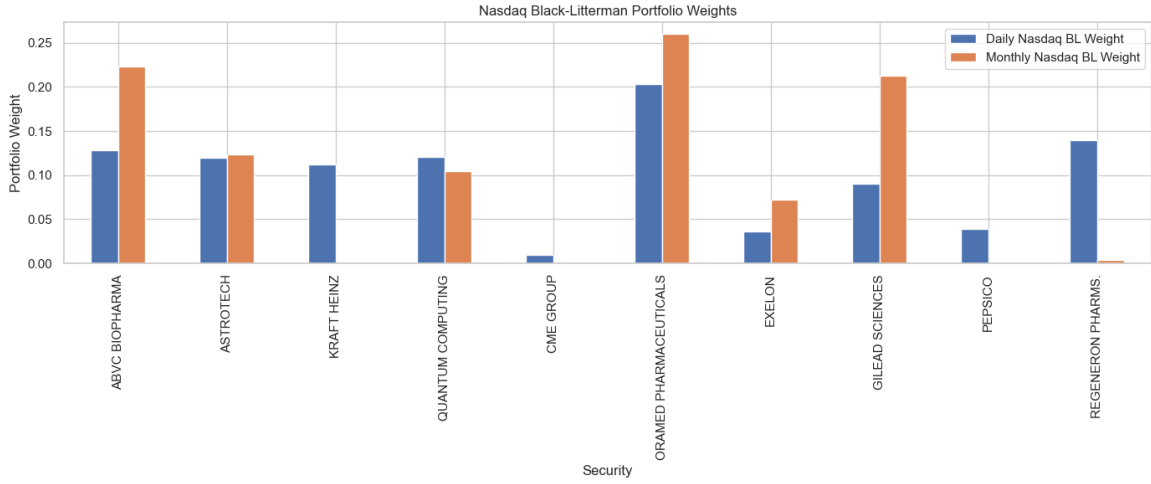


Figure 18: Nasdaq Black-Litterman Portfolio Wreights

Metric	Daily BL	Monthly BL
Mean	0.061%	0.709%
Std	2.124%	11.200%
Variance	0.000451	0.012544
Skewness	0.672	0.374
Excess Kurtosis	10.051	−0.293
Sharpe Ratio	0.025	0.048

Table 22: Nasdaq Black-Litterman Portfolio Statistics

Comparing with the long-only MV portfolio: the BL daily Sharpe (0.025) is lower than the MV long-only (0.058), and the BL monthly Sharpe is substantially lower than the MV long-only (0.271). Here too, we carry out a comparison with the long-only MV portfolio, but the results are not particularly worth noting as they are identical to the previous case for the NYSE, given that this pattern is typical of the BL approach, which allows for more diversified and robust portfolios through the use of priors, thereby limiting the role of historical return estimates, albeit at the cost of lowering the estimated Sharpe ratio.

12 Q22&Q23 — Global Minimum Variance Portfolio: NYSE and Nasdaq

The Global Minimum Variance (GMV) portfolio is the type of allocation that allows to minimizes the total portfolio variance, without any constraint on expected return.

Unlike the mean-variance optimal portfolio, which requires an estimate of expected returns, the GMV portfolio depends only on the covariance matrix, making it more robust to estimation error in the return vector. The GMV optimization problem is:

$$\min_w w^\top \Sigma w \quad \text{subject to} \quad \sum_{i=1}^N w_i = 1, \quad w_i \geq 0 \quad \forall i$$

The long-only constraint is maintained for consistency with the portfolio construction framework of the preceding sections.

Security	Daily GMV Weight	Monthly GMV Weight
TEAM	0.0115	0.0037
Kroger	0.1537	0.2137
FTI Consulting	0.1548	0.1701
McCormick & Co	0.1761	0.1519
Nordic Amer. Tankers	0.0412	0.0242
Humana	0.0887	0.1318
Telephone & Data Sys.	0.0142	0.0014
Eli Lilly	0.1142	0.1470
Northrop Grumman	0.1640	0.1562
Newmont	0.0815	0.0000

Table 23: NYSE Global Minimum Variance Portfolio Weights

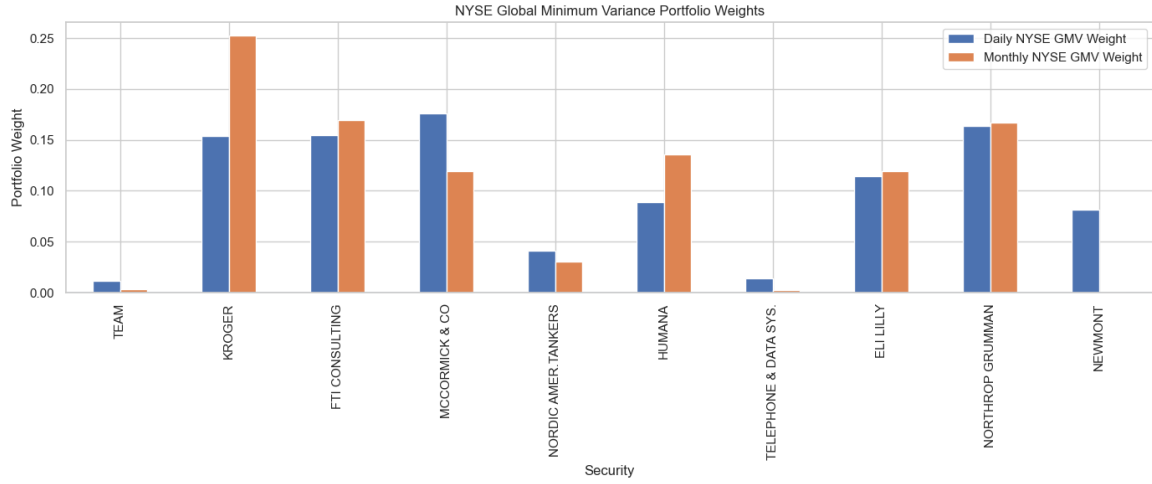


Figure 19: NYSE Global Minimum Variance Portfolio Weights

Looking at the portfolio weights, we see once again that all ten securities selected at the outset have a non-zero weight, with a very similar balance among the four most significant securities: with McCormick & Co, Northrop Grumman, Kroger and FTI Consulting. In this way we are able to adequately represent the volatility profile of the NYSE, as no single security is significantly more volatile than the others; consequently, the optimizer exploits diversification benefits across the full set rather than concentrating in a handful of names. It should be noted that Newmont receives a positive, albeit small, weighting at daily intervals, whilst it is the only security not to receive one when switching to monthly intervals. This suggests that the contribution of this security to the portfolio is to be found in the short term, where its gold-driven return dynamics are more independent of the other securities.

Metric	Daily GMV	Monthly GMV
Mean	0.039%	0.850%
Std	0.869%	3.358%
Variance	0.000076	0.001128
Skewness	0.484	0.259
Excess Kurtosis	4.878	-0.146
Sharpe Ratio	0.036	0.207

Table 24: **NYSE Global Minimum Variance Portfolio Statistics**

Once again, the Sharpe ratio has fallen, both on a daily and monthly basis, compared with the long-only MV portfolio. However, it is worth noting that they represent the minimum-risk allocation for an investor who is unwilling to make any assumption about expected returns.

The excess kurtosis of the daily GMV confirms the presence of fat tails even in the minimum-variance allocation, driven by the idiosyncratic events of the individual securities. Monthly aggregation reduces excess kurtosis to a slightly negative value, indicating a platykurtic distribution with thinner tails than a normal distribution, consistent with the diversification benefits of holding a broadly spread portfolio over longer horizons.

As regards the Nasdaq GMV, not all securities are assigned weights, even when using daily rather than monthly data. Quantum Computing, for instance, is excluded from the selection in the daily data set but is assigned weights in the monthly data set; this is most likely due to its very low correlation with the other securities, as seen above. The dominant holdings are Exelon, CME Group, PepsiCo, Kraft Heinz, and Gilead Sciences, which are all defensive securities. This is the direct output of the variance minimization objective: the optimizer identifies the subset of securities with the lowest individual volatilities and the most favorable covariance structure, which in the Nasdaq universe are precisely the consumer staples, utilities, and established pharmaceutical names rather than the speculative technology names.

At monthly frequency, the allocation shifts substantially. CME Group becomes the dominant holding followed by PepsiCo, Kraft Heinz. These three alone already account for 72% of the portfolio's total weighting, whilst Gilead Sciences and Oramed Pharmaceuticals drop to zero weight.

Security	Daily GMV Weight	Monthly GMV Weight
ABVC Biopharma	0.0066	0.0003
Astrotech	0.0273	0.0722
Kraft Heinz	0.1088	0.1656
Quantum Computing	0.0017	0.0225
CME Group	0.2839	0.3298
Oramed Pharmaceuticals	0.0083	0.0000
Exelon	0.1610	0.0474
Gilead Sciences	0.1137	0.0000
PepsiCo	0.2092	0.2418
Regeneron Pharmaceuticals	0.0795	0.1204

Table 25: **Nasdaq Global Minimum Variance Portfolio Weights**

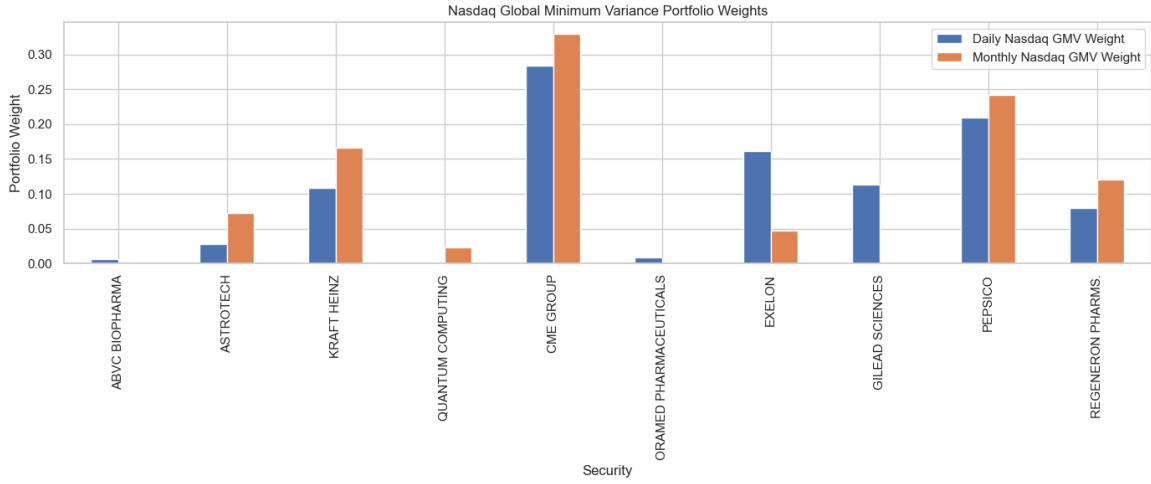


Figure 20: Nasdaq Global Minimum Variance Portfolio Wreights

The Nasdaq GMV Sharpe Ratios are very low, reflecting the fundamental challenge of the Nasdaq portfolio, which is that even the minimum-variance allocation carries meaningful volatility because all selected securities are inherently more volatile than typical NYSE names.

Metric	Daily GMV	Monthly GMV
Mean	0.020%	0.205%
Std	0.807%	3.197%
Variance	0.000065	0.001022
Skewness	−0.200	−0.317
Excess Kurtosis	2.009	−0.334
Sharpe Ratio	0.015	0.012

Table 26: Nasdaq Global Minimum Variance Portfolio Statistics

Ultimately, the portfolio built on the NYSE consistently outperforms its counterpart based on Nasdaq securities, regardless of the frequency considered. This confirms that the NYSE universe, even at its minimum-risk point, offers a more favorable risk-return trade-off.

Both GMV portfolios achieve lower Sharpe Ratios than their respective long-only MV portfolios, as foreseeable by construction, since the GMV minimizes variance without regard for return, while the MV portfolio explicitly maximizes the Sharpe Ratio.

13 Q24 — Hybrid Portfolio: Linear Combination of Allocation Models

The four portfolio construction methods developed in the preceding sections have different set of assumptions about expected returns, risk, and estimation uncertainty. Due to this, they produce meaningfully different allocations. Consequently, in this final section, we aim to identify the source of these differences and then construct a hybrid model that combines the approaches discussed by an equal-weighted linear combination of the four models.

Mean-Variance portfolio is highly sensitive to the estimated expected return vector, treats historical sample means as exact inputs and allocates aggressively to whichever securities have the most favorable recent return-to-risk ratio. This results in a portfolio that is theoretically optimal within the training sample but fragile when tested on new data, highlighting a phenomenon known as the maximization problem

The Black-Litterman portfolio replaces the historical mean with a posterior expected return that blends a prior equilibrium allocation with subjective investor views. The prior anchors the allocation toward a diversified baseline and prevents the optimizer from taking extreme positions based on noisy historical estimates. The result is a more broadly diversified portfolio, but one whose properties depend on the quality and consistency of the views specified.

The Pure Bayesian portfolio follows a similar logic but derives the posterior mean and covariance directly, without requiring explicit views. It shrinks the sample mean toward a prior grand mean and adjusts the covariance accordingly. In the present implementation, because the prior mean is set to the cross-sectional average of sample returns, the Pure Bayesian portfolio tends to produce allocations very similar to the MV long-only portfolio.

The Global Minimum Variance portfolio ignores expected returns entirely and focuses solely on minimizing variance. It is the most robust to estimation error in the return vector, but it sacrifices any exposure to high-return assets, resulting in the lowest Sharpe Ratio among the four models in most specifications.

The hybrid portfolio is constructed as follows:

$$w_{Hybrid} = 0.25 w_{MV} + 0.25 w_{BL} + 0.25 w_{Bayesian} + 0.25 w_{GMV}$$

where each component receives equal weight. The resulting weights are normalized to enforce the full-investment constraint. We have chosen to give equal weighting to all the models because, at this stage, we have no idea which one will prove to be the best; consequently, this uncertainty has led us to adopt a cautious approach whilst also exploring all possible methodologies.

Security	MV	BL	Bayesian	GMV	Hybrid
TEAM	0.0000	0.1033	0.0000	0.0115	0.0287
Kroger	0.2137	0.2849	0.2138	0.1537	0.2165
FTI Consulting	0.0000	0.0686	0.0000	0.1548	0.0559
McCormick & Co	0.0000	0.1192	0.0000	0.1761	0.0738
Nordic Amer. Tankers	0.1049	0.0248	0.1048	0.0412	0.0689
Humana	0.0000	0.0813	0.0000	0.0887	0.0425
Telephone & Data Sys.	0.0553	0.1239	0.0553	0.0142	0.0622
Eli Lilly	0.5070	0.0657	0.5069	0.1142	0.2984
Northrop Grumman	0.0442	0.0438	0.0443	0.1640	0.0741
Newmont	0.0749	0.0844	0.0749	0.0815	0.0789

Table 27: **NYSE Portfolio Weights Across Models — Daily**

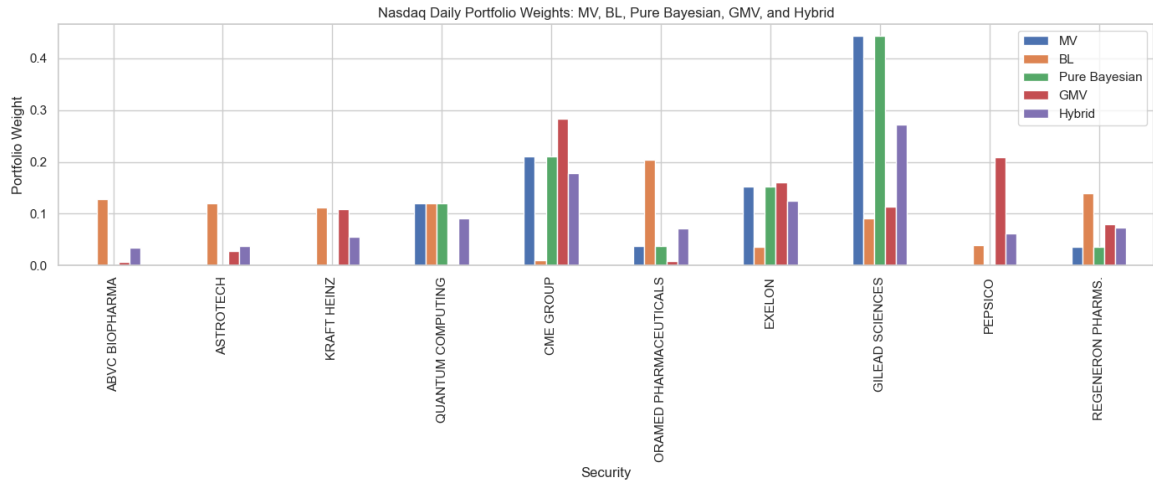


Figure 21: NYSE Hybrid Portfolio Werights — Daily

Security	MV	BL	Bayesian	GMV	Hybrid
TEAM	0.0035	0.0786	0.0035	0.0037	0.0223
Kroger	0.1757	0.3403	0.1787	0.2527	0.2368
FTI Consulting	0.0000	0.0395	0.0000	0.1698	0.0523
McCormick & Co	0.0000	0.1291	0.0000	0.1194	0.0621
Nordic Amer. Tankers	0.0561	0.0261	0.0561	0.0305	0.0422
Humana	0.0091	0.0236	0.0114	0.1356	0.0449
Telephone & Data Sys.	0.0321	0.1481	0.0318	0.0023	0.0536
Eli Lilly	0.4864	0.0421	0.4811	0.1193	0.2822
Northrop Grumman	0.1934	0.0000	0.1943	0.1668	0.1386
Newmont	0.0436	0.1727	0.0430	0.0000	0.0648

Table 28: NYSE Portfolio Weights Across Models — Monthly

	Pure Bayesian	MV	Hybrid	GMV	BL
Mean	0.001071	0.001071	0.000757	0.000390	0.000496
Std	0.012611	0.012613	0.010126	0.008690	0.011697
Variance	0.000159	0.000159	0.000103	0.000076	0.000137
Skewness	0.500388	0.500524	0.305094	−0.142259	0.802591
Excess Kurtosis	5.635429	5.637102	3.303349	2.354323	8.199139
Sharpe Ratio	0.078604	0.078604	0.066924	0.035789	0.035651

Table 29: NYSE Daily — All-Model Statistics Ranked by Sharpe Ratio

	MV	Pure Bayesian	Hybrid	GMV	BL
Mean	0.021214	0.021080	0.015609	0.008557	0.011587
Std	0.049152	0.048816	0.041197	0.033305	0.060707
Variance	0.002416	0.002383	0.001697	0.001109	0.003685
Skewness	0.217091	0.223564	0.424555	0.433068	0.767060
Excess Kurtosis	0.574324	0.589265	1.262459	0.393223	1.519366
Sharpe Ratio	0.397687	0.397674	0.338434	0.206880	0.163411

Table 30: NYSE Monthly — All-Model Statistics Ranked by Sharpe Ratio

Security	MV	BL	Bayesian	GMV	Hybrid
ABVC Biopharma	0.0000	0.1280	0.0000	0.0066	0.0336
Astrotech	0.0000	0.1202	0.0000	0.0273	0.0369
Kraft Heinz	0.0000	0.1121	0.0000	0.1088	0.0552
Quantum Computing	0.1207	0.1204	0.1206	0.0017	0.0909
CME Group	0.2109	0.0100	0.2111	0.2839	0.1790
Oramed Pharmaceuticals	0.0374	0.2035	0.0374	0.0083	0.0717
Exelon	0.1522	0.0363	0.1516	0.1610	0.1253
Gilead Sciences	0.4433	0.0908	0.4434	0.1137	0.2728
PepsiCo	0.0000	0.0391	0.0000	0.2092	0.0621
Regeneron Pharmaceuticals	0.0355	0.1398	0.0359	0.0795	0.0727

Table 31: Nasdaq Portfolio Weights Across Models — Daily

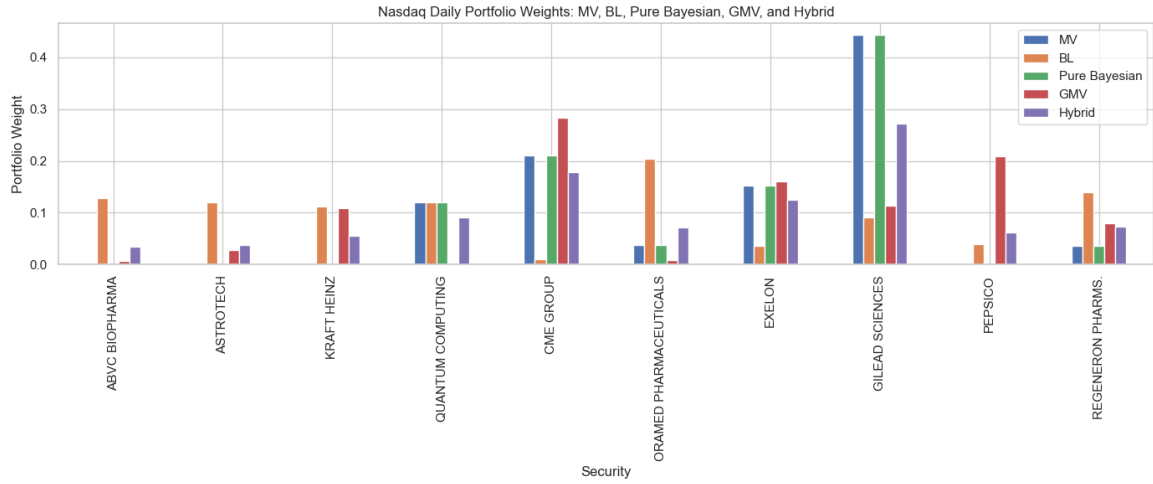


Figure 22: Nasdaq Hybrid Portfolio Wreights

Security	MV	BL	Bayesian	GMV	Hybrid
ABVC Biopharma	0.0000	0.2236	0.0000	0.0003	0.0560
Astrotech	0.0000	0.1236	0.0000	0.0722	0.0490
Kraft Heinz	0.0000	0.0000	0.0000	0.1656	0.0414
Quantum Computing	0.0843	0.1041	0.0835	0.0225	0.0736
CME Group	0.1646	0.0000	0.1670	0.3298	0.1653
Oramed Pharmaceuticals	0.0000	0.2601	0.0000	0.0000	0.0650
Exelon	0.1467	0.0720	0.1487	0.0474	0.1037
Gilead Sciences	0.2528	0.2127	0.2507	0.0000	0.1790
PepsiCo	0.0000	0.0000	0.0000	0.2418	0.0605
Regeneron Pharmaceuticals	0.3516	0.0038	0.3501	0.1204	0.2065

Table 32: Nasdaq Portfolio Weights Across Models — Monthly

	Pure Bayesian	MV	Hybrid	BL	GMV
Mean	0.000893	0.000893	0.000647	0.000606	0.000198
Std	0.013940	0.013942	0.012439	0.021238	0.008071
Variance	0.000194	0.000194	0.000155	0.000451	0.000065
Skewness	1.017947	1.018636	0.533959	0.672427	-0.200261
Excess Kurtosis	5.562234	5.567160	3.598483	10.093871	2.021104
Sharpe Ratio	0.058346	0.058346	0.045671	0.024803	0.014738

Table 33: **Nasdaq Daily — All-Model Statistics Ranked by Sharpe Ratio**

	MV	Pure Bayesian	Hybrid	BL	GMV
Mean	0.015938	0.015856	0.010234	0.007086	0.002058
Std	0.052655	0.052353	0.052789	0.112000	0.031971
Variance	0.002773	0.002741	0.002787	0.012544	0.001022
Skewness	0.447068	0.430588	0.325857	0.383923	-0.324579
Excess Kurtosis	1.026591	0.967072	0.280998	-0.211754	-0.256287
Sharpe Ratio	0.271027	0.271021	0.162303	0.048390	0.012247

Table 34: **Nasdaq Monthly — All-Model Statistics Ranked by Sharpe Ratio**

Several observations emerge from the comparison tables.

First of all, we can say that, in both markets and for both frequency the MV long-only and Pure Bayesian portfolios produce almost exactly the same weights and statistics. This is a consequence of the prior specification in the Bayesian model: setting the prior mean equal to the cross-sectional grand mean of sample returns produces a posterior that is very close to the sample mean when n_{obs} is large relative to $\kappa_0 = 1$. With 1305 daily observations, the sample information dominates completely and the posterior converges to the sample mean. In conclusion, we can therefore say that the Pure Bayesian strategy offers very few diversification benefits compared to the MV in the implementation we have examined.

In both markets, the BL portfolio is the only one that assigns meaningful positive weight to securities that receive zero weight in MV and Bayesian, TEAM, FTI Consulting, McCormick, and Humana in the NYSE case and ABVC Biopharma, Astrotech, and Kraft Heinz in the Nasdaq case. This is the primary value added by the BL model: by anchoring to an equal-weight prior, it prevents the optimizer from zeroing out securities that happen to have unfavorable recent return estimates.

Across all four specifications, the GMV portfolio achieves the lowest standard deviation but also the lowest Sharpe Ratio. As we mentioned in our previous analyses, this was predictable from a theoretical perspective given the structure of the model, which minimises variance without taking returns into account. The result is, in fact, a portfolio that is as safe as possible, but the cost is that it loses a great deal of efficiency.

Finally, looking at the hybrid model we have just constructed, we can see that, as might be expected given the equal use of weights, it falls somewhere between the four models mentioned. The hybrid volatility is lower than MV because the GMV and BL components pull weight toward less volatile securities, and this pattern is shown at each frequency and portfolio.

What can be gleaned from this model is that by combining the strengths and weaknesses of the previous models, we obtain a model which, whilst it is true that it does not have the highest Sharpe Ratio, is robust against any potential model failure, mitigated by such an even distribution of weights.

This robustness is useful because it leads us to believe with complete certainty that, in out-of-sample results, the hybrid model is likely to have a significant advantage over the others.

14 Introduction and Motivation

14.1 The Baseline: Bansal and Yaron (2004)

The natural reference point for this literature is Bansal-Yaron (2004)?. Their model builds on two ingredients. First, aggregate consumption contains a small but highly persistent expected-growth component x_t — a “long-run risk” that is nearly invisible at short horizons but compounds substantially over time. Second, investors have Epstein-Zin recursive preferences, which separate risk aversion γ from the elasticity of intertemporal substitution (EIS) ψ . This separation matters: under the standard calibration

$$\gamma > 1 > \frac{1}{\psi}, \tag{1}$$

agents have a strict preference for the early resolution of uncertainty. They dislike being exposed to long-horizon consumption risk and are willing to pay to insure against it.

The empirical payoff is substantial. Under this calibration the model accounts for the high equity premium, the low risk-free rate, and countercyclical price-dividend ratios; something that standard power-utility models struggle to replicate. For two decades, this framework has been the workhorse of equilibrium asset pricing.

14.2 Why Bansal–Yaron Fails on Term Structures

The model’s difficulties emerge once one looks beyond unconditional moments. Beeler-Campbell (2012) and van Binsbergen et al. (2012) document two failures of the standard calibration, both concerning the term structure of risk premia.

Failure 1: Bond risk premia. Empirically, the average yield curve slopes upward for both nominal and real bonds (Figure 1 of Doh and Wu (2017)): investors require higher expected returns for holding longer-maturity debt. The standard Bansal–Yaron calibration implies the

opposite. Because agents are so eager to insure against long-run consumption risk, long-term real bonds act as valuable hedges, and their risk premia turn negative. The model produces a downward-sloping term structure of real bond risk premia, a direct, unavoidable consequence of the preference for early resolution.

Failure 2: Equity risk premia on dividend strips. van Binsbergen et al. (2012) extract returns on dividend strips, claims to the dividend paid exactly n periods from now, from options data and find a downward-sloping term structure: short-horizon strips earn higher risk premia than long-horizon ones, and Sharpe ratios decline with maturity. The standard calibration again implies the opposite. Because agents dislike long-run dividend risk, they demand higher compensation for long-duration equity claims. Both the risk premium and the Sharpe ratio are increasing in maturity. Both are counterfactual.

The two failures are not independent. They share the same root cause: investor preferences for early resolution of uncertainty. The same mechanism that makes long bonds valuable insurance (driving their premia down) also makes long dividend strips expensive to hold (driving their premia up). Any model that tries to fix one failure with this preference channel automatically worsens the other.

Several papers attempt partial remedies. Belo et al. (2015) introduce a time-varying, procyclical leverage ratio q_t to make dividends riskier in the short run, which can generate a downward-sloping equity term structure while preserving the standard preference calibration but does not change the bond risk-premium implications at all. Marfè (2016) adds wage insurance to create short-run dividend volatility and even obtains an upward-sloping bond term structure, but neither of these papers can generate a downward-sloping term structure of Sharpe ratios. The common shortcoming is that none of these approaches break the underlying preference mechanism.

14.3 What Doh and Wu (2017) Add

Doh and Yu (2017) extend the standard long-run risks framework along two dimensions that, taken together, break the structural deadlock described above.

The first is a quadratic log wealth-consumption ratio. Rather than assuming a linear (affine) relation between $\log(W_t/C_t)$ and X_t , the model postulates

$$z_{c,t} = \lambda_{c,0} + \lambda'_{c,1}X_t + \frac{1}{2}X'_tH_cX_t, \quad (2)$$

with the symmetric matrix H_c left free. Following Le-Singleton (2010) and Doh-Yu(2015), this generates a time-varying market price of risk: the sensitivity of the pricing kernel to risk factors now depends on X_t itself. In the standard linear model, by contrast, the market price of risk is constant, and risk premia are flat across maturities at their unconditional mean.

The second is a reverse-engineering of expected cash flows. The standard approach specifies g_c , g_d , g_π exogenously and solves the Euler equations for the price coefficients. This path, with a quadratic $z_{c,t}$, requires solving a matrix quadratic equation in H_c for which no closed-

form solution exists in the multi-factor case. Doh and Wu reverse the logic: they postulate the quadratic price functions and derive the implied expected cash flows directly from the Euler equation restrictions. The coefficients then follow in closed form, without any numerical procedure.

The economic mechanism is straightforward. When $H_c \neq 0$, the market price of risk acquires a quadratic term in X_t , opening a second channel through which shocks affect risk premia, one that operates through the variance of the pricing kernel rather than only through its covariance with cash flows. Because $\lambda_{c,1}$ and H_c can be chosen independently of γ and ψ , the model can simultaneously deliver:

- an upward-sloping average term structure of bond risk premia;
- a downward-sloping average term structure of equity risk premia;
- a downward-sloping term structure of Sharpe ratios on dividend strips.

All three obtain under the standard preference calibration, preference for early resolution of uncertainty, $\psi > 1$, without any tension between them. The remainder of this paper derives all analytical results of Doh-Wu (2017) step by step.

15 Model Setup

15.1 State Variables

The economy is driven by a 2×1 Markovian state vector $X_t \in \mathbb{R}^2$ evolving as:

$$\Delta X_{t+1} = X_{t+1} - X_t = \Phi X_t + \Sigma_x \varepsilon_{x,t+1}, \quad \varepsilon_{x,t+1} \sim \mathcal{N}(0, I), \quad \text{i.i.d.} \quad (3)$$

Φ is a 2×2 mean-reversion matrix; stationarity requires all eigenvalues of $I + \Phi$ to lie strictly inside the unit circle. We write $\Omega_x = \Sigma_x \Sigma'_x$ and assume it invertible.

15.2 Cash Flows and Inflation

Consumption growth, dividend growth, and inflation are each decomposed into a predictable component and an i.i.d. shock:

$$\Delta c_{t+1} = g_c(X_t) + \sigma_c \varepsilon_{c,t+1}, \quad (4)$$

$$\Delta d_{t+1} = g_d(X_t) + \rho_{cd} \varepsilon_{c,t+1} + \sigma_d \varepsilon_{d,t+1}, \quad (5)$$

$$\pi_{t+1} = g_\pi(X_t) + \sigma_\pi \varepsilon_{\pi,t+1}. \quad (6)$$

The shocks $\varepsilon_c, \varepsilon_d, \varepsilon_\pi$ are mutually independent $\mathcal{N}(0, 1)$.

Stacking into $Y_{t+1} = (\Delta X'_{t+1}, \Delta c_{t+1}, \Delta d_{t+1}, \pi_{t+1})'$, the joint covariance matrix is

$$\Omega = \begin{pmatrix} \Omega_x & 0 & 0 & 0 \\ 0 & \sigma_c^2 & \rho_{cd}^2 \sigma_c^2 & 0 \\ 0 & \rho_{cd}^2 \sigma_c^2 & \rho_{cd}^2 \sigma_c^2 + \sigma_d^2 & 0 \\ 0 & 0 & 0 & \sigma_\pi^2 \end{pmatrix}. \quad (7)$$

The block-diagonal structure separates persistent shocks, which are transmitted through X_t , from transitory ones, and is the key identifying assumption of the model.

15.3 Investor Preferences: Epstein–Zin

Investors have recursive preferences (Epstein and Zin, 1989):

$$U_t = \left[(1 - \delta) C_t^{1-1/\psi} + \delta (\mathbb{E}_t[U_{t+1}^{1-\gamma}])^{1/\theta} \right]^{1/(1-1/\psi)}, \quad \theta = \frac{1 - \gamma}{1 - 1/\psi}. \quad (8)$$

From Epstein–Zin (1991), the log real stochastic discount factor is

$$-m_{t+1} = -\theta \ln \delta + \frac{\theta}{\psi} \Delta c_{t+1} - (\theta - 1) r_{c,t+1}, \quad (9)$$

where $r_{c,t+1}$ is the log real return on the consumption claim (the asset paying C_t each period). In Section 17 we'll derive the explicit CDF, showing how it is possible that the coefficient of Δc_{t+1} is γ . Section 17.

For assets with nominal payoffs, the log nominal SDF is

$$m_{t+1}^n = m_{t+1} - \pi_{t+1}. \quad (10)$$

Agents prefer early resolution of uncertainty when $\gamma > 1/\psi$, i.e. $\theta < 0$. The *standard calibration* $\gamma > 1 > 1/\psi$ ensures this while allowing a high EIS.

16 Asset Prices and the Reverse-Engineering Strategy

16.1 The Reverse-Engineering Idea

Standard approach specifies $g_c(X_t)$ exogenously and solves the Euler equation $\mathbb{E}_t[e^{m_{t+1} + r_{c,t+1}}] = 1$ for the coefficients of $z_{c,t} = \log(W_t/C_t)$. With time-varying market prices of risk (i.e. $z_{c,t}$ quadratic in X_t), this requires solving a matrix quadratic equation in H_c :

$$\frac{\theta^2 k_{c,2}^2}{2} \sigma_x^2 H_c^2 + \theta (2k_{c,2} \Phi - k_{c,1}) H_c + (1 - \gamma) W_c = 0, \quad (11)$$

for which no closed-form solution exists in the multi-factor case (see Appendix A.1 of Doh and Wu).

Doh and Wu reverse the logic: they postulate quadratic functional forms for the price ratios and recover the implied expected cash flows from the Euler equation restrictions. All coefficients follow in closed form, so it's not required a numerical procedure, and that's the real advantage of the model.

16.2 The Three Posted Asset Prices

Consumption claim. We postulate that the log wealth-consumption ratio is a quadratic function of the state vector:

$$z_{c,t} = f_c(X_t) = \lambda_{c,0} + \lambda'_{c,1}X_t + \frac{1}{2}X'_t H_c X_t, \quad (12)$$

with $\bar{z}_c = \lambda_{c,0}$ the steady-state value, $\lambda_{c,1} \in \mathbb{R}^2$, and H_c a symmetric 2×2 matrix. When $\bar{H}_c \neq 0$ the market price of risk becomes state-dependent, as shown in Section 17.

To obtain a tractable return expression, we apply the Campbell–Shiller (1989) log-linearization. The gross return is defined as:

$$R_{c,t+1} = \frac{P_{c,t+1} + C_{t+1}}{P_{c,t}}, \quad (13)$$

where $P_{c,t}$ is the ex-dividend price of the consumption claim at time t . Multiplying and dividing by C_{t+1} in the numerator and C_t in the denominator, this rewrites as:

$$R_{c,t+1} = \frac{P_{c,t+1}/C_{t+1} + 1}{P_{c,t}/C_t} \cdot \frac{C_{t+1}}{C_t}. \quad (14)$$

Taking logs and using $z_{c,t} = \ln(P_{c,t}/C_t)$:

$$r_{c,t+1} = \ln(e^{z_{c,t+1}} + 1) - z_{c,t} + \Delta c_{t+1}. \quad (15)$$

Since $\ln(e^z + 1)$ is nonlinear in $z_{c,t+1}$, we linearize around $\bar{z}_c$ via a first-order Taylor expansion:

$$\begin{aligned} \ln(e^{z_{c,t+1}} + 1) &\approx \ln(1 + e^{\bar{z}_c}) + \frac{e^{\bar{z}_c}}{1 + e^{\bar{z}_c}} (z_{c,t+1} - \bar{z}_c) \\ &= \underbrace{\ln(1 + e^{\bar{z}_c}) - \bar{z}_c \frac{e^{\bar{z}_c}}{1 + e^{\bar{z}_c}}}_{k_{c,0}} + \underbrace{\frac{e^{\bar{z}_c}}{1 + e^{\bar{z}_c}}}_{k_{c,2}} z_{c,t+1}. \end{aligned} \quad (16)$$

Substituting into (15) and writing $z_{c,t+1} = z_{c,t} + \Delta z_{c,t+1}$:

$$r_{c,t+1} \approx k_{c,0} - \underbrace{(1 - k_{c,2})}_{k_{c,1}} z_{c,t} + k_{c,2} \Delta z_{c,t+1} + \Delta c_{t+1}, \quad (17)$$

where the three constants depend only on $\bar{z}_c = \lambda_{c,0}$:

$$k_{c,2} = \frac{e^{\bar{z}_c}}{1 + e^{\bar{z}_c}}, \quad k_{c,1} = 1 - k_{c,2}, \quad k_{c,0} = \ln(1 + e^{\bar{z}_c}) - \bar{z}_c k_{c,2}. \quad (18)$$

Note that $k_{c,1}, k_{c,2} \in (0, 1)$ with $k_{c,1} + k_{c,2} = 1$, so the approximation is a convex combination of the current and future log wealth-consumption ratio. The quality of this approximation improves as Ω_x is small relative to the quadratic coefficients in $z_{c,t}$, exactly the calibration regime of long-run risks models.

To handle $\Delta z_{c,t+1}$, Le and Singleton (2010) show that the first-order Taylor approximation

$$\Delta f_c(X_{t+1}) \approx \Gamma_c(X_t)' \Delta X_{t+1}, \quad \Gamma_c(X_t) = \frac{\partial f_c}{\partial X_t} = \lambda_{c,1} + \bar{H}_c X_t, \quad \bar{H}_c = \frac{H_c + H'_c}{2}, \quad (19)$$

is accurate when Ω_x is small relative to the quadratic coefficients. To derive the gradient explicitly: with $X_t = (x_1, x_2)'$,

$$\frac{\partial}{\partial X_t} \left(\frac{1}{2} X_t' H_c X_t \right) = \frac{1}{2} \frac{\partial}{\partial X_t} (H_{11} x_1^2 + (H_{12} + H_{21}) x_1 x_2 + H_{22} x_2^2) = \frac{H_c + H'_c}{2} X_t = \bar{H}_c X_t.$$

Substituting (19) and (3) into (17):

$$r_{c,t+1} \approx \underbrace{k_{c,0} - k_{c,1} f_c(X_t) + k_{c,2} \Gamma_c(X_t)' \Phi X_t + g_c(X_t)}_{\mathbb{E}_t(r_{c,t+1})} + \underbrace{k_{c,2} \Gamma_c(X_t)' \Sigma_x \varepsilon_{x,t+1} + \sigma_c \varepsilon_{c,t+1}}_{\text{unexpected part}}. \quad (20)$$

The market price of risk vector $h_c(X_t)$ (loadings on $Y_{t+1} - \mathbb{E}_t(Y_{t+1})$) is therefore:

$$h_c(X_t) = [k_{c,2} \Gamma_c(X_t)', 1, 0, 0]'. \quad (21)$$

Since $\Gamma_c(X_t) = \lambda_{c,1} + \bar{H}_c X_t$, when $\bar{H}_c \neq 0$ the market price of risk is time-varying and state-dependent.

To see explicitly why $h_c(X_t)$ takes this form, note that the unexpected part of $r_{c,t+1}$ from equation (20) is:

$$r_{c,t+1} - \mathbb{E}_t(r_{c,t+1}) = k_{c,2} \Gamma_c(X_t)' \Sigma_x \varepsilon_{x,t+1} + \sigma_c \varepsilon_{c,t+1}.$$

The full shock vector is $Y_{t+1} - \mathbb{E}_t(Y_{t+1}) = (\Sigma_x \varepsilon'_{x,t+1}, \sigma_c \varepsilon_{c,t+1}, \sigma_d \varepsilon_{d,t+1}, \sigma_\pi \varepsilon_{\pi,t+1})'$. Writing the unexpected return as $h_c(X_t)'(Y_{t+1} - \mathbb{E}_t(Y_{t+1}))$ and matching block by block:

$$\begin{aligned} \text{block } \Sigma_x \varepsilon_{x,t+1} : & \quad k_{c,2} \Gamma_c(X_t), \\ \text{block } \sigma_c \varepsilon_{c,t+1} : & \quad 1, \\ \text{block } \sigma_d \varepsilon_{d,t+1} : & \quad 0 \quad (r_{c,t+1} \text{ does not depend on dividend shocks}), \\ \text{block } \sigma_\pi \varepsilon_{\pi,t+1} : & \quad 0 \quad (r_{c,t+1} \text{ does not depend on inflation shocks}). \end{aligned}$$

Hence $h_c(X_t) = [k_{c,2} \Gamma_c(X_t)', 1, 0, 0]'$, confirming equation (21).

One-period dividend strip.

$$z_{d,t}^1 = f_d^1(X_t) = \lambda_{d,0} + \lambda'_{d,1} X_t + \frac{1}{2} X_t' H_d X_t. \quad (22)$$

The return on a one-period strip is:

$$r_{d,t+1}^1 = \Delta d_{t+1} - f_d^1(X_t). \quad (23)$$

Derivation: the strip price is $P_{1,t}^d = D_t e^{z_{d,t}^1}$; its payoff at $t + 1$ is D_{t+1} . Hence $r_{d,t+1}^1 = \ln(D_{t+1}/P_{1,t}^d) = \Delta d_{t+1} - z_{d,t}^1$.

Nominal short rate via Taylor rule.

$$i_{1,t} = \bar{r} + \alpha_c g_c(X_t) + \alpha_\pi g_\pi(X_t) + i'_m X_t, \quad (24)$$

where $\alpha_\pi > 1$ (Taylor principle) ensures determinacy of inflation, and $i_m = (0, 1)'$ selects the second state variable as a nominal shock.

17 The Stochastic Discount Factor in Explicit Form

Substituting (20) into (9) and using $\Delta c_{t+1} = g_c(X_t) + \sigma_c \varepsilon_{c,t+1}$:

$$\begin{aligned} -m_{t+1} = & -\theta \ln \delta - (\theta - 1)k_{c,0} + (\theta - 1)k_{c,1}f_c(X_t) \\ & - (\theta - 1)k_{c,2}\Gamma_c(X_t)'\Delta X_{t+1} + \gamma\Delta c_{t+1}. \end{aligned} \quad (25)$$

Substituting $\Delta X_{t+1} = \Phi X_t + \Sigma_x \varepsilon_{x,t+1}$ and $\Delta c_{t+1} = g_c(X_t) + \sigma_c \varepsilon_{c,t+1}$, and expanding $f_c(X_t) = \lambda_{c,0} + \lambda'_{c,1}X_t + \frac{1}{2}X_t'H_cX_t$ and $g_c(X_t) = g_{c,0} + g'_{c,1}X_t + \frac{1}{2}X_t'W_cX_t$, the full expression becomes:

$$\begin{aligned} -m_{t+1} = & \underbrace{-\theta \ln \delta - (\theta - 1)(k_{c,0} - k_{c,1}\lambda_{c,0} + \gamma g_{c,0})}_{m_0, \text{ purely constant}} \\ & + \underbrace{(\theta - 1)\left[k_{c,1}\lambda'_{c,1}X_t + \frac{k_{c,1}}{2}X_t'H_cX_t - k_{c,2}\Gamma_c(X_t)'\Phi X_t\right] + \gamma\left[g'_{c,1}X_t + \frac{1}{2}X_t'W_cX_t\right]}_{m_1(X_t), \text{ function of } X_t \text{ only}} \\ & + \underbrace{-(\theta - 1)k_{c,2}\Gamma_c(X_t)'\Sigma_x \varepsilon_{x,t+1} + \gamma\sigma_c \varepsilon_{c,t+1}}_{\text{unexpected part, zero mean}}. \end{aligned}$$

m_0 is a purely deterministic constant, capturing the average level of the log pricing kernel and depending on the time discount factor δ , the steady-state wealth-consumption ratio (through $k_{c,0}$ and $\lambda_{c,0}$), and the average consumption growth $g_{c,0}$ but not on X_t or with any shock:

$$m_0 = -\theta \ln \delta - (\theta - 1)[k_{c,0} - k_{c,1}\lambda_{c,0} + \gamma g_{c,0}]. \quad (26)$$

$m_1(X_t)$ is the predictable, state-dependent component of the SDF which is known at time t given X_t , but varies over time as X_t moves. It captures how the expected level of the pricing kernel responds to the current state of the economy. It has three contributions:

- $(\theta - 1)k_{c,1}[\lambda'_{c,1}X_t + \frac{1}{2}X_t'H_cX_t]$: the effect of the current wealth-consumption ratio $f_c(X_t)$ on the SDF, scaled by $(\theta - 1)$ on $r_{c,t+1}$.

- $-(\theta - 1)k_{c,2}\Gamma_c(X_t)'\Phi X_t$: the effect of expected future changes in the wealth-consumption ratio (through $\mathbb{E}_t[\Delta z_{c,t+1}] = \Gamma_c(X_t)'\Phi X_t$)
- $\gamma[g'_{c,1}X_t + \frac{1}{2}X_t'W_cX_t]$: the effect of expected consumption growth $g_c(X_t)$, scaled by risk aversion γ .

Formally:

$$m_1(X_t) = (\theta - 1) \left[k_{c,1}\lambda'_{c,1}X_t + \frac{k_{c,1}}{2}X_t'H_cX_t - k_{c,2}\Gamma_c(X_t)'\Phi X_t \right] + \gamma \left[g'_{c,1}X_t + \frac{1}{2}X_t'W_cX_t \right]. \quad (27)$$

Deriving the market price of risk $h(X_t)$. The unexpected part of $-m_{t+1}$ is:

$$-m_{t+1} - \mathbb{E}_t(-m_{t+1}) = -(\theta - 1)k_{c,2}\Gamma_c(X_t)'\Sigma_x\varepsilon_{x,t+1} + \gamma\sigma_c\varepsilon_{c,t+1}.$$

We write this as $h(X_t)'(Y_{t+1} - \mathbb{E}_t(Y_{t+1}))$ and match block by block against $Y_{t+1} - \mathbb{E}_t(Y_{t+1}) = (\Sigma_x\varepsilon'_{x,t+1}, \sigma_c\varepsilon_{c,t+1}, \sigma_d\varepsilon_{d,t+1}, \sigma_\pi\varepsilon_{\pi,t+1})'$:

$$\begin{aligned} \text{block } \Sigma_x\varepsilon_{x,t+1} : & \quad -(\theta - 1)k_{c,2}\Gamma_c(X_t) = (1 - \theta)k_{c,2}\Gamma_c(X_t), \\ \text{block } \sigma_c\varepsilon_{c,t+1} : & \quad \gamma, \\ \text{block } \sigma_d\varepsilon_{d,t+1} : & \quad 0 \quad (m_{t+1} \text{ does not load on dividend shocks}), \\ \text{block } \sigma_\pi\varepsilon_{\pi,t+1} : & \quad 0 \quad (m_{t+1} \text{ does not load on inflation shocks}). \end{aligned}$$

Hence:

$$h(X_t) = [(1 - \theta)k_{c,2}\Gamma_c(X_t)', \gamma, 0, 0]'. \quad (28)$$

The SDF therefore decomposes cleanly into its expected and unexpected parts:

$$-m_{t+1} = \underbrace{m_0 + m_1(X_t)}_{\mathbb{E}_t(-m_{t+1})} + \underbrace{h(X_t)'(Y_{t+1} - \mathbb{E}_t(Y_{t+1}))}_{\text{zero-mean shock}}. \quad (29)$$

Since $\Gamma_c(X_t) = \lambda_{c,1} + \bar{H}_cX_t$, when $\bar{H}_c \neq 0$ the vector $h(X_t)$ depends on X_t : the market price of risk is time-varying. In the linear case ($\bar{H}_c = 0$) it collapses to the constant vector $[(1 - \theta)k_{c,2}\lambda'_{c,1}, \gamma, 0, 0]'$, generating flat risk premia across maturities.

18 Determining Expected Cash Flows via Euler Equations

18.1 The Log-Normal Euler Equation

In equilibrium every asset i satisfies $\mathbb{E}_t[e^{m_{t+1}+r_{i,t+1}}] = 1$. Since $m_{t+1} + r_{i,t+1}$ is conditionally Gaussian given X_t , taking logs of both sides gives:

$$\mathbb{E}_t(m_{t+1} + r_{i,t+1}) + \frac{1}{2}\text{Var}_t(m_{t+1} + r_{i,t+1}) = 0. \quad (30)$$

Because the left-hand side is a quadratic polynomial in X_t , equation (30) must hold for every value of X_t . We therefore match coefficients of degree 0, 1, and 2 separately.

18.2 Determining $g_c(X_t)$

18.2.1 Computing $\mathbb{E}_t(m_{t+1} + r_{c,t+1})$

Using (20) and (29), and the fact that the unexpected part has zero mean:

$$\mathbb{E}_t(m_{t+1} + r_{c,t+1}) = -m_0 - m_1(X_t) + \mathbb{E}_t(r_{c,t+1}).$$

We expand each component by degree in X_t .

Expansion of $-m_0$ (degree 0):

$$-m_0 = \theta \ln \delta + (\theta - 1)[k_{c,0} - k_{c,1}\lambda_{c,0} + \gamma g_{c,0}].$$

Expansion of $-m_1(X_t)$. Recall $\Gamma_c(X_t)' \Phi X_t = \lambda'_{c,1} \Phi X_t + X_t' \bar{H}'_c \Phi X_t$.

Degree 1 from $-m_1$:

$$-(\theta - 1)k_{c,1}\lambda'_{c,1}X_t + (\theta - 1)k_{c,2}\lambda'_{c,1}\Phi X_t - \gamma g'_{c,1}X_t.$$

Degree 2 from $-m_1$:

$$-\frac{(\theta - 1)k_{c,1}}{2}X_t'H_cX_t + (\theta - 1)k_{c,2}X_t'\bar{H}'_c\Phi X_t - \frac{\gamma}{2}X_t'W_cX_t.$$

Expansion of $\mathbb{E}_t(r_{c,t+1})$. From (20), using $f_c(X_t) = \lambda_{c,0} + \lambda'_{c,1}X_t + \frac{1}{2}X_t'H_cX_t$:

Degree 0:

$$k_{c,0} - k_{c,1}\lambda_{c,0} + g_{c,0}.$$

Degree 1:

$$-k_{c,1}\lambda'_{c,1}X_t + k_{c,2}\lambda'_{c,1}\Phi X_t + g'_{c,1}X_t.$$

Degree 2:

$$-\frac{k_{c,1}}{2}X_t'H_cX_t + k_{c,2}X_t'\bar{H}'_c\Phi X_t + \frac{1}{2}X_t'W_cX_t.$$

Collecting by degree. We sum contributions from $-m_0$, $-m_1(X_t)$, and $\mathbb{E}_t(r_{c,t+1})$; the variance term $\frac{1}{2}\text{Var}_t(m_{t+1} + r_{c,t+1})$ is added in Section 18.2.2.

Degree 0:

$$\theta \ln \delta + \theta(k_{c,0} - k_{c,1}\lambda_{c,0}) + [(\theta - 1)\gamma + 1]g_{c,0}. \quad (31)$$

Degree 1. Three groups of terms contribute:

$$\begin{aligned}\lambda'_{c,1} \text{ terms: } & [-(\theta - 1)k_{c,1} - k_{c,1}] \lambda'_{c,1} = -\theta k_{c,1} \lambda'_{c,1}, \\ \lambda'_{c,1} \Phi \text{ terms: } & [(\theta - 1)k_{c,2} + k_{c,2}] \lambda'_{c,1} \Phi = +\theta k_{c,2} \lambda'_{c,1} \Phi, \\ g_{c,1} \text{ terms: } & (-\gamma + 1) g_{c,1} = (1 - \gamma) g_{c,1}.\end{aligned}$$

Collecting:

$$\theta [-k_{c,1} \lambda'_{c,1} + k_{c,2} \lambda'_{c,1} \Phi] X_t + (1 - \gamma) g'_{c,1} X_t. \quad (32)$$

Degree 2. Three groups of terms contribute:

$$\begin{aligned}H_c \text{ terms: } & [-(\theta - 1) - 1] \frac{k_{c,1}}{2} = -\frac{\theta k_{c,1}}{2}, \\ \bar{H}'_c \Phi \text{ terms: } & [(\theta - 1) + 1] k_{c,2} = +\theta k_{c,2}, \\ W_c \text{ terms: } & \left(-\frac{\gamma}{2} + \frac{1}{2}\right) = \frac{1 - \gamma}{2}.\end{aligned}$$

Collecting:

$$\theta \left[-\frac{k_{c,1}}{2} X'_t H_c X_t + k_{c,2} X'_t \bar{H}'_c \Phi X_t \right] + \frac{1 - \gamma}{2} X'_t W_c X_t. \quad (33)$$

18.2.2 Computing $\frac{1}{2} \text{Var}_t(m_{t+1} + r_{c,t+1})$

The variance depends entirely on its unexpected component. The unexpected part of $r_{c,t+1}$ is $h_c(X_t)'(Y_{t+1} - \mathbb{E}_t(Y_{t+1}))$ from equation (21), while the unexpected part of m_{t+1} is $-h(X_t)'(Y_{t+1} - \mathbb{E}_t(Y_{t+1}))$ from equation (29). The sum therefore has loadings $(h_c(X_t) - h(X_t))'$. Subtracting (28) from (21):

$$h_c(X_t) - h(X_t) = \begin{pmatrix} k_{c,2} \Gamma_c \\ 1 \\ 0 \\ 0 \end{pmatrix} - \begin{pmatrix} (1 - \theta) k_{c,2} \Gamma_c \\ \gamma \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} \theta k_{c,2} \Gamma_c(X_t) \\ 1 - \gamma \\ 0 \\ 0 \end{pmatrix}.$$

Using the block-diagonal structure of Ω (equation (7)), the zero entries for dividends and inflation contribute nothing, so:

$$\text{Var}_t(m_{t+1} + r_{c,t+1}) = \theta^2 k_{c,2}^2 \Gamma_c(X_t)' \Omega_x \Gamma_c(X_t) + (1 - \gamma)^2 \sigma_c^2.$$

Expanding $\Gamma_c(X_t) = \lambda_{c,1} + \bar{H}_c X_t$:

$$\frac{1}{2} \text{Var}_t = \underbrace{\frac{\theta^2 k_{c,2}^2}{2} \lambda'_{c,1} \Omega_x \lambda_{c,1} + \frac{(1 - \gamma)^2 \sigma_c^2}{2}}_{\text{degree 0}} + \underbrace{\theta^2 k_{c,2}^2 \lambda'_{c,1} \Omega_x \bar{H}_c X_t}_{\text{degree 1}} + \underbrace{\frac{\theta^2 k_{c,2}^2}{2} X'_t \bar{H}'_c \Omega_x \bar{H}_c X_t}_{\text{degree 2}}. \quad (34)$$

18.2.3 Matching Coefficients to Obtain g_c

Adding (31)–(33) and (34), and imposing that each degree equals zero:

Degree 0.

$$\theta \ln \delta + \theta(k_{c,0} - k_{c,1}\lambda_{c,0}) + [(\theta - 1)\gamma + 1]g_{c,0} + \frac{\theta^2 k_{c,2}^2}{2}\lambda'_{c,1}\Omega_x \lambda_{c,1} + \frac{(1 - \gamma)^2 \sigma_c^2}{2} = 0.$$

Using $\theta = (1 - \gamma)/(1 - 1/\psi)$ and solving for $g_{c,0}$:

$$g_{c,0} = -\frac{1}{1 - 1/\psi}[\ln \delta + k_{c,0} - k_{c,1}\lambda_{c,0}] - \frac{1 - \gamma}{2}\sigma_c^2 - \frac{\theta^2 k_{c,2}^2}{2(1 - \gamma)}\lambda'_{c,1}\Omega_x \lambda_{c,1}. \quad (35)$$

Degree 1. Setting the coefficient of X_t to zero:

$$\theta[-k_{c,1}I + k_{c,2}\Phi']\lambda_{c,1} + (1 - \gamma)g_{c,1} + \theta^2 k_{c,2}^2 \bar{H}_c \Omega_x \lambda_{c,1} = 0.$$

Dividing by $-(1 - \gamma)$ and using $\theta^2/(1 - \gamma) = \theta/(1 - 1/\psi)$:

$$g_{c,1} = \frac{1}{1 - 1/\psi}[k_{c,1}I - k_{c,2}\Phi']\lambda_{c,1} - \frac{\theta^2 k_{c,2}^2}{1 - \gamma}\bar{H}_c \Omega_x \lambda_{c,1}. \quad (36)$$

Degree 2. Setting the coefficient of $X_t' \cdot X_t$ to zero:

$$\theta\left[-\frac{k_{c,1}}{2}H_c + k_{c,2}\bar{H}_c'\Phi\right] + \frac{1 - \gamma}{2}W_c + \frac{\theta^2 k_{c,2}^2}{2}\bar{H}_c \Omega_x \bar{H}_c = 0.$$

Solving for W_c :

$$W_c = \frac{1}{1 - 1/\psi}(k_{c,1}H_c - 2k_{c,2}\bar{H}_c\Phi) - \frac{\theta^2 k_{c,2}^2}{1 - \gamma}\bar{H}_c \Omega_x \bar{H}_c. \quad (37)$$

Note that $W_c \neq 0$ if and only if $\bar{H}_c \neq 0$. This last term is the curvature term, which makes expected consumption growth quadratic in X_t and allows to generate time-varying market prices of risk.

In summary:

$$g_c(X_t) = g_{c,0} + g'_{c,1}X_t + \frac{1}{2}X_t'W_cX_t, \quad (38)$$

with $g_{c,0}$, $g_{c,1}$, W_c given by (35)–(37).

18.3 Determining $g_d(X_t)$

Applying (30) to the one-period strip return $r_{d,t+1}^1 = \Delta d_{t+1} - z_{d,t}^1$:

$$\mathbb{E}_t(m_{t+1} + \Delta d_{t+1} - z_{d,t}^1) + \frac{1}{2}\text{Var}_t(m_{t+1} + \Delta d_{t+1} - z_{d,t}^1) = 0.$$

The unexpected component of the strip return is $h'_d(Y_{t+1} - \mathbb{E}_t(Y_{t+1}))$ with $h_d = [0, \rho_{cd}, 1, 0]'$, so:

$$h_d - h(X_t) = \begin{pmatrix} -(1 - \theta)k_{c,2}\Gamma_c(X_t) \\ \rho_{cd} - \gamma \\ 1 \\ 0 \end{pmatrix}.$$

The variance is therefore:

$$\text{Var}_t = (1 - \theta)^2 k_{c,2}^2 \Gamma_c(X_t)' \Omega_x \Gamma_c(X_t) + (\rho_{cd} - \gamma)^2 \sigma_c^2 + \sigma_d^2.$$

Matching coefficients after expanding by degree in X_t (the algebra proceeds identically to the g_c case, replacing $h_c - h$ by $h_d - h$):

$$g_d(X_t) = g_{d,0} + g'_{d,1}X_t + \frac{1}{2}X_t'W_dX_t, \quad (39)$$

where

$$g_{d,0} = \lambda_{d,0} + \gamma g_{c,0} - \theta \ln \delta - (\theta - 1)(k_{c,0} - k_{c,1}\lambda_{c,0}) - \frac{k_{c,2}^2(\theta - 1)^2 \lambda'_{c,1} \Omega_x \lambda_{c,1} + (\rho_{cd} - \gamma)^2 \sigma_c^2 + \sigma_d^2}{2}, \quad (40)$$

$$g_{d,1} = \lambda_{d,1} + \gamma g_{c,1} + (\theta - 1)(k_{c,1}I - k_{c,2}\Phi')\lambda_{c,1}, \quad (41)$$

$$W_d = H_d + \gamma W_c + (\theta - 1)k_{c,1}H_c - 2(\theta - 1)k_{c,2}\bar{H}_c\Phi - k_{c,2}^2(\theta - 1)^2\bar{H}_c\Omega_x\bar{H}_c. \quad (42)$$

Notice that $g_{d,1}$ (equation (41)) contains no term involving $\bar{H}_c\Omega_x\lambda_{c,1}$, unlike $g_{c,1}$ (equation (36)). The reason is a sign difference: in $h_d - h$ the coefficient on Γ_c is $-(1 - \theta) = (\theta - 1)$, whereas in $h_c - h$ it is $+\theta k_{c,2}$. As a result, the cross-term $2(\theta - 1)k_{c,2}\bar{H}_c\Omega_x\lambda_{c,1}$ that would appear at degree 1 is instead absorbed into the degree-2 matching via W_d , and cancels at the linear level when combined with the expected-return terms.

Each coefficient of g_d inherits the structure of the corresponding coefficient of g_c , scaled by γ plus corrections proportional to $(\theta - 1)$, the signature of the EIS–risk-aversion separation.

18.4 Determining $g_\pi(X_t)$

The nominal short rate can be pinned down in two ways: directly from the Taylor rule (24), or from the Euler equation for the nominal one-period bond ($P_{1,t}^\$ = e^{-i_{1,t}}$), giving $\mathbb{E}_t[e^{m_{t+1}^n + i_{1,t}}] = 1$, or in log-normal form:

$$i_{1,t} = -\mathbb{E}_t(m_{t+1}^n) - \frac{1}{2}\text{Var}_t(m_{t+1}^n).$$

18.4.1 Computing $\mathbb{E}_t(m_{t+1}^n)$

Since $\mathbb{E}_t(\pi_{t+1}) = g_\pi(X_t)$ by definition and $\mathbb{E}_t(m_{t+1}) = -m_0 - m_1(X_t)$ from the decomposition (29):

$$\mathbb{E}_t(m_{t+1}^n) = \mathbb{E}_t(m_{t+1}) - \mathbb{E}_t(\pi_{t+1}) = -m_0 - m_1(X_t) - g_\pi(X_t).$$

18.4.2 Computing $\frac{1}{2}\text{Var}_t(m_{t+1}^n)$

The unexpected part of $m_{t+1}^n = m_{t+1} - \pi_{t+1}$ is $-(h(X_t) + h_\pi)'(Y_{t+1} - \mathbb{E}_t(Y_{t+1}))$ where $h_\pi = [0, 0, 0, 1]'$. Thus:

$$h(X_t) + h_\pi = [(1 - \theta)k_{c,2}\Gamma_c(X_t)', \gamma, 0, 1]',$$

and (using block-diagonal Ω):

$$\text{Var}_t(m_{t+1}^n) = (1 - \theta)^2 k_{c,2}^2 \Gamma_c(X_t)' \Omega_x \Gamma_c(X_t) + \gamma^2 \sigma_c^2 + \sigma_\pi^2.$$

18.4.3 Equating and Solving for g_π

Equating the two expressions for $i(1, t)$ and isolating g_π :

$$(\alpha_\pi - 1)g_\pi(X_t) = m_0 + m_1(X_t) - \alpha_c g_c(X_t) - i'_m X_t - \bar{r} - \frac{1}{2}\text{Var}_t(m_{t+1}^n).$$

The denominator $(\alpha_\pi - 1) > 0$ by the Taylor principle. The matching by degree proceeds identically to Section 5.2, replacing $h_c - h$ by $h(X_t) + h_\pi$ throughout. We therefore report only the final coefficients:

$$g_\pi(X_t) = g_{\pi,0} + g'_{\pi,1} X_t + \frac{1}{2} X_t' W_\pi X_t, \quad (43)$$

where

$$g_{\pi,0} = \frac{-\bar{r} - \theta \ln \delta + (1 - \theta)(k_{c,0} - k_{c,1}\lambda_{c,0}) + (\gamma - \alpha_c)g_{c,0} - \frac{1}{2} [\gamma^2 \sigma_c^2 + \sigma_\pi^2 + k_{c,2}^2 (\theta - 1)^2 \lambda'_{c,1} \Omega_x \lambda_{c,1}]}{\alpha_\pi - 1}, \quad (44)$$

$$g_{\pi,1} = \frac{(\theta - 1)(k_{c,1}I - k_{c,2}\Phi)' \lambda_{c,1} + (\gamma - \alpha_c)g_{c,1} - (\theta - 1)^2 k_{c,2}^2 \bar{H}_c \Omega_x \lambda_{c,1} - i_m}{\alpha_\pi - 1}, \quad (45)$$

$$W_\pi = \frac{(\theta - 1)(k_{c,1}H_c - 2k_{c,2}\bar{H}_c\Phi) + (\gamma - \alpha_c)W_c - (\theta - 1)^2 k_{c,2}^2 \bar{H}_c \Omega_x \bar{H}_c}{\alpha_\pi - 1}. \quad (46)$$

All nine coefficients $(g_{c,0}, g_{c,1}, W_c, g_{d,0}, g_{d,1}, W_d, g_{\pi,0}, g_{\pi,1}, W_\pi)$ are uniquely determined in closed form by the free parameters $(\lambda_{c,0}, \lambda_{c,1}, H_c, \lambda_{d,0}, \lambda_{d,1}, H_d, \alpha_c, \alpha_\pi)$. So, there's no quadratic matrix to solve, and this is the core gain of the reverse-engineering approach.

19 Term Structure of Interest Rates and Bond Risk Premia

19.1 Nominal Term Structure

19.1.1 The Nominal Short Rate

Substituting the quadratic forms (38) and (43) into the Taylor rule (24):

$$i_{1,t} = A_{\$,1} + B'_{\$,1}X_t + \frac{1}{2}X'_tC_{\$,1}X_t, \quad (47)$$

with

$$A_{\$,1} = \bar{r} + \alpha_c g_{c,0} + \alpha_\pi g_{\pi,0}, \quad B_{\$,1} = \alpha_c g_{c,1} + \alpha_\pi g_{\pi,1} + i_m, \quad C_{\$,1} = \alpha_c W_c + \alpha_\pi W_\pi.$$

19.1.2 Pricing the Nominal n -Period Bond

Let $P_{n,t}^\$$ be the nominal price of a zero-coupon bond paying \$1 at $t + n$. The no-arbitrage recursion is:

$$P_{n,t}^\$ = E_t[e^{m_{t+1}^n} P_{n-1,t+1}^\$]. \quad (48)$$

Conjecture:

$$P_{n,t}^\$ = e^{-A_{\$,n} - B'_{\$,n}X_t - \frac{1}{2}X'_tC_{\$,n}X_t}. \quad (49)$$

Taking logs of the conjectured $n-1$ price and substituting $X_{t+1} = (I + \Phi)X_t + \Sigma_x \varepsilon_{x,t+1}$:

$$-B'_{\$,n-1}X_{t+1} = -B'_{\$,n-1}(I + \Phi)X_t - B'_{\$,n-1}\Sigma_x \varepsilon_{x,t+1}.$$

For the quadratic term we apply the Le-Singleton approximation (19). Since $X'_{t+1}C_{\$,n-1}X_{t+1}$ is nonlinear in $\varepsilon_{x,t+1}$, we linearise around $\varepsilon_{x,t+1} = 0$, i.e. around $X_{t+1} = (I + \Phi)X_t$. Using $X'CX = X'\bar{C}X$ for any matrix C , with $\bar{C}_{\$,n-1} = \frac{1}{2}(C_{\$,n-1} + C'_{\$,n-1})$:

$$-\frac{1}{2}X'_{t+1}C_{\$,n-1}X_{t+1} \approx -\frac{1}{2}X'_tC_{\$,n-1}X_t - X'_t\bar{C}_{\$,n-1}\Phi X_t - X'_t\bar{C}_{\$,n-1}\Sigma_x \varepsilon_{x,t+1}.$$

Collecting, $\ln P_{n-1,t+1}^\$$ decomposes into an expected part and a shock part:

$$\begin{aligned} \ln P_{n-1,t+1}^\$ = & \underbrace{-A_{\$,n-1} - B'_{\$,n-1}(I + \Phi)X_t - \frac{1}{2}X'_tC_{\$,n-1}X_t - X'_t\bar{C}_{\$,n-1}\Phi X_t}_{E_t[\ln P_{n-1,t+1}^\$]} \\ & - \underbrace{(B_{\$,n-1} + 2\bar{C}_{\$,n-1}X_t)' \Sigma_x \varepsilon_{x,t+1}}_{\text{shock on } X\text{-block}}. \end{aligned}$$

The full unexpected part of $m_{t+1}^n + \ln P_{n-1,t+1}^\$$ is $-(h(X_t) + \tilde{h}_{\$,n-1}(X_t))'(Y_{t+1} - E_t Y_{t+1})$. We

read off $\tilde{h}_{\$,n-1}$ block by block against $Y_{t+1} - E_t Y_{t+1} = (\Sigma_x \varepsilon'_{x,t+1}, \sigma_c \varepsilon_{c,t+1}, \sigma_d \varepsilon_{d,t+1}, \sigma_\pi \varepsilon_{\pi,t+1})'$:

$$\tilde{h}_{\$,n-1}(X_t) = \begin{pmatrix} B_{\$,n-1} + 2\bar{C}_{\$,n-1}X_t \\ 0 \\ 0 \\ 1 \end{pmatrix}.$$

The loading $B_{\$,n-1} + 2\bar{C}_{\$,n-1}X_t$ on the $\Sigma_x \varepsilon_{x,t+1}$ block comes directly from differentiating $\ln P_{n-1,t+1}^\$$ with respect to $\varepsilon_{x,t+1}$. The entry 1 on the $\sigma_\pi \varepsilon_{\pi,t+1}$ block comes from the nominal SDF: $m_{t+1}^n = m_{t+1} - \pi_{t+1}$, so the inflation shock enters $-(h + \tilde{h})$ with coefficient -1 , i.e. on $\tilde{h}$ with $+1$. The dividend block is zero because neither m_{t+1}^n nor $\ln P_{n-1,t+1}^\$$ loads on $\sigma_d \varepsilon_{d,t+1}$.

The total loading vector is therefore:

$$h(X_t) + \tilde{h}_{\$,n-1}(X_t) = \begin{pmatrix} (1 - \theta)k_{c,2}\Gamma_c(X_t) + B_{\$,n-1} + 2\bar{C}_{\$,n-1}X_t \\ \gamma \\ 0 \\ 1 \end{pmatrix}. \quad (50)$$

Computing the variance. Using the block-diagonal structure of Ω (7) and expanding $\Gamma_c(X_t) = \lambda_{c,1} + \bar{H}_c X_t$:

$$\frac{1}{2}\text{Var}_t[m_{t+1}^n + \ln P_{n-1,t+1}^\$] = \frac{1}{2}\|(1 - \theta)k_{c,2}\Gamma_c(X_t) + B_{\$,n-1} + 2\bar{C}_{\$,n-1}X_t\|_{\Omega_x}^2 + \frac{1}{2}\gamma^2\sigma_c^2 + \frac{1}{2}\sigma_\pi^2.$$

Expanding the squared norm, the terms $\frac{1}{2}(1 - \theta)^2 k_{c,2}^2 \|\Gamma_c\|_{\Omega_x}^2 + \frac{1}{2}\gamma^2\sigma_c^2 + \frac{1}{2}\sigma_\pi^2$ are already absorbed in $A_{\$,1}$ by construction of $i_{1,t}$ and cancel in the matching step. The remaining cross-terms between $h(X_t)$ and $\tilde{h}_{\$,n-1}(X_t)$, and the pure $\tilde{h}$ -squared terms, contribute by degree:

$$\text{degree 0: } (1 - \theta)k_{c,2}\lambda'_{c,1}\Omega_x B_{\$,n-1} + \frac{1}{2}B'_{\$,n-1}\Omega_x B_{\$,n-1}, \quad (51)$$

$$\text{degree 1: } (1 - \theta)k_{c,2}\left[X'_t \bar{H}'_c \Omega_x B_{\$,n-1} + B'_{\$,n-1}\Omega_x \bar{C}_{\$,n-1}X_t\right] + 2B'_{\$,n-1}\Omega_x \bar{C}_{\$,n-1}X_t, \quad (52)$$

$$\text{degree 2: } 2(1 - \theta)k_{c,2}X'_t \bar{H}'_c \Omega_x \bar{C}_{\$,n-1}X_t + 2X'_t \bar{C}'_{\$,n-1}\Omega_x \bar{C}_{\$,n-1}X_t. \quad (53)$$

Applying $\ln E_t(e^Z) = E_t(Z) + \frac{1}{2}\text{Var}_t(Z)$ and equating to $-A_{\$,n} - B'_{\$,n}X_t - \frac{1}{2}X'_t C'_{\$,n}X_t$:

Degree 0. The constant part of the expected sum contributes $-A_{\$,n-1} - A_{\$,1}$ (where $-A_{\$,1}$ collects $-m_0 - g_{\pi,0}$ by definition of $i_{1,t}$). Adding (51) and using $(1 - \theta) = -(\theta - 1)$:

$$A_{\$,n} = A_{\$,n-1} + A_{\$,1} + (\theta - 1)k_{c,2}\lambda'_{c,1}\Omega_x B_{\$,n-1} - \frac{1}{2}B'_{\$,n-1}\Omega_x B_{\$,n-1}. \quad (54)$$

The term $(\theta - 1)k_{c,2}\lambda'_{c,1}\Omega_x B_{\$,n-1}$ is the covariance between the constant part of the pricing kernel and the bond's exposure to X -shocks; it is the fundamental source of the average bond risk premium. The term $-\frac{1}{2}B'_{\$,n-1}\Omega_x B_{\$,n-1}$ is the Jensen correction, not a risk premium.

Degree 1. The linear part of the expected sum contributes $-B'_{\$,n-1}(I + \Phi)$ from the bond price,

minus $B'_{\$,1}$ from $i_{1,t}$. Adding (52) and symmetrising:

$$B_{\$,n} = B_{\$,1} + (\theta - 1)k_{c,2}(\bar{H}'_c \Omega_x B_{\$,n-1} + \bar{C}'_{\$,n-1} \Omega_x \lambda_{c,1}) + (I + \Phi - \Omega_x \bar{C}_{\$,n-1})' B_{\$,n-1}. \quad (55)$$

Two terms enter through $(\theta - 1)k_{c,2}$. The first, $\bar{H}'_c \Omega_x B_{\$,n-1}$, comes from the state-dependent part of $h(X_t)$ crossing with the constant loading $B_{\$,n-1}$ of $\tilde{h}$. When $\bar{H}_c = 0$ it vanishes, and is therefore absent in the linear model. The second, $\bar{C}'_{\$,n-1} \Omega_x \lambda_{c,1}$, reflects how the bond's price sensitivity interacts with the base market price of risk, and is present even in the linear model.

When $\bar{H}_c = 0$ both channels reduce to a maturity-independent constant and risk premia are flat across maturities.

Degree 2. The quadratic part of the expected sum contributes $-2\bar{C}_{\$,n-1}\Phi$ from the bond price, minus $C_{\$,1}$ from $i_{1,t}$. Adding (53):

$$C_{\$,n} = C_{\$,n-1} + C_{\$,1} + 2\bar{C}_{\$,n-1}\Phi + 2(\theta - 1)k_{c,2}\bar{H}_c \Omega_x \bar{C}_{\$,n-1} - \bar{C}_{\$,n-1} \Omega_x \bar{C}_{\$,n-1}, \quad (56)$$

with $\bar{C}_{\$,n-1} = \frac{1}{2}(C_{\$,n-1} + C'_{\$,n-1})$. The term $-\bar{C}_{\$,n-1} \Omega_x \bar{C}_{\$,n-1}$ is negative semi-definite: it is the Jensen correction at degree 2, penalizing bondholders for convexity. Its sign will be reversed for dividend strips in Section 19.3, where strip holders benefit from convexity.

The nominal yield is $i_{n,t} = (A_{\$,n} + B'_{\$,n}X_t + \frac{1}{2}X'_t C_{\$,n}X_t)/n$.

19.1.3 Nominal Bond Risk Premium

The one-period expected excess return of an n -period nominal bond is defined as:

$$rp_{n,t}^{\$} = E_t[\ln P_{n-1,t+1}^{\$} - \ln P_{n,t}^{\$} - i_{1,t}].$$

Using the Le-Singleton approximation for $E_t[\ln P_{n-1,t+1}^{\$}]$:

$$E_t[\ln P_{n-1,t+1}^{\$}] \approx -A_{\$,n-1} - B'_{\$,n-1}(I + \Phi)X_t - \frac{1}{2}X'_t C_{\$,n-1}X_t - X'_t \bar{C}_{\$,n-1}\Phi X_t.$$

Subtracting the logarithm of (49) and (47) and collecting by degree (noting that $-X'_t \bar{C}_{\$,n-1}\Phi X_t = -\frac{1}{2}X'_t(\bar{C}_{\$,n-1}\Phi + \Phi'\bar{C}_{\$,n-1})X_t$ after symmetrisation):

$$\begin{aligned} rp_{n,t}^{\$} \approx & \underbrace{A_{\$,n} - A_{\$,n-1} - A_{\$,1}}_{\text{constant}} + \underbrace{(B_{\$,n} - B_{\$,n-1}(I + \Phi) - B_{\$,1})' X_t}_{\text{linear}} \\ & + \underbrace{\frac{X'_t(C_{\$,n} - C_{\$,n-1} - C_{\$,1} - 2\bar{C}_{\$,n-1}\Phi)X_t}{2}}_{\text{quadratic}}. \end{aligned} \quad (57)$$

In the linear model ($C_{\$,n} = 0$, $\bar{H}_c = 0$, $\mathbb{E}(X_t) = 0$) only the constant term survives: bond risk premia are time-invariant. In the quadratic model, the linear and quadratic terms in X_t generate time-varying risk premia.

19.2 Real Term Structure

19.2.1 The Real Short Rate

From the Euler equation for the real one-period bond, $e^{-r_{1,t}} = E_t(e^{m_{t+1}})$, so $r_{1,t} = -E_t(m_{t+1}) - \frac{1}{2}\text{Var}_t(m_{t+1})$. The unexpected part of m_{t+1} is $-h(X_t)'(Y_{t+1} - E_t Y_{t+1})$ from (29), so:

$$\text{Var}_t(m_{t+1}) = (1 - \theta)^2 k_{c,2}^2 \Gamma_c(X_t)' \Omega_x \Gamma_c(X_t) + \gamma^2 \sigma_c^2.$$

Expanding by degree in X_t :

$$r_{1,t} = A_1 + B_1' X_t + \frac{1}{2} X_t' C_1 X_t, \quad (58)$$

with

$$A_1 = -\theta \ln \delta + \gamma g_{c,0} + (1 - \theta)(k_{c,0} - k_{c,1} \lambda_{c,0}) - \frac{\gamma^2 \sigma_c^2 + (1 - \theta)^2 k_{c,2}^2 \lambda_{c,1}' \Omega_x \lambda_{c,1}}{2}, \quad (59)$$

$$B_1 = \gamma g_{c,1} + (\theta - 1)(k_{c,1} - k_{c,2} \Phi') \lambda_{c,1} - (\theta - 1)^2 k_{c,2}^2 \bar{H}_c \Omega_x \lambda_{c,1}, \quad (60)$$

$$C_1 = \gamma W_c + (\theta - 1) k_{c,1} \bar{H}_c - 2(\theta - 1) k_{c,2} \bar{H}_c \Phi - (\theta - 1)^2 k_{c,2}^2 \bar{H}_c \Omega_x \bar{H}_c. \quad (61)$$

19.2.2 Pricing the Real n -Period Bond

With conjecture $P_{n,t} = e^{-A_n - B_n' X_t - \frac{1}{2} X_t' C_n X_t}$ and no-arbitrage condition $P_{n,t} = E_t(e^{m_{t+1}} P_{n-1,t+1})$, the derivation proceeds identically to Section 6.1.2, replacing the nominal SDF $m_{t+1}^n = m_{t+1} - \pi_{t+1}$ with the real SDF m_{t+1} . Without the inflation loading, the σ_π^2 term and the $h_\pi = [0, 0, 0, 1]'$ cross-terms are absent, and the $\bar{H}_c$ cross-term feeds directly into the coefficient on B_{n-1} , rather than outside the bracket. The recursions for $n \geq 2$ are:

$$A_n = A_{n-1} + A_1 + (\theta - 1) k_{c,2} \lambda_{c,1}' \Omega_x B_{n-1} - \frac{1}{2} B_{n-1}' \Omega_x B_{n-1}, \quad (62)$$

$$B_n = \left(I + \Phi - \Omega_x (\bar{C}_{n-1} - (\theta - 1) k_{c,2} \bar{H}_c) \right)' B_{n-1} + B_1 + (\theta - 1) k_{c,2} \bar{C}_{n-1} \Omega_x \lambda_{c,1}, \quad (63)$$

$$C_n = C_{n-1} + C_1 + 2 \bar{C}_{n-1} \Phi - \bar{C}_{n-1} \Omega_x \bar{C}_{n-1} + 2(\theta - 1) k_{c,2} \bar{H}_c \Omega_x \bar{C}_{n-1}. \quad (64)$$

The real yield is $r_{n,t} = (A_n + B_n' X_t + \frac{1}{2} X_t' C_n X_t)/n$.

19.2.3 Real Bond Risk Premium

By the same argument as Section 6.1.3:

$$rp_{n,t} \approx (A_n - A_{n-1} - A_1) + (B_n - B_{n-1}(I + \Phi) - B_1)' X_t + \frac{X_t'(C_n - C_{n-1} - C_1 - 2\bar{C}_{n-1}\Phi)X_t}{2}. \quad (65)$$

The structure is identical to the nominal case (57). The difference in term structures between real and nominal bonds arises entirely from the different recursions (62)–(64) vs. (54)–(56): the nominal recursions incorporate the inflation variance σ_π^2 and the covariance between inflation and the pricing kernel through $h(X_t) + h_\pi$.

19.3 The Term Structure of Dividend Strips and Equity Risk Premia

19.3.1 Pricing the Dividend Strip

Let $P_{n,t}^d$ be the real price of a claim to dividend D_{t+n} . Define $z_{n,t}^d = \ln(P_{n,t}^d/D_t)$. The pricing recursion is:

$$e^{z_{n,t}^d} = E_t \left[e^{\Delta d_{t+1} + m_{t+1} + z_{n-1,t+1}^d} \right]. \quad (66)$$

The one-period case was postulated: $z_{1,t}^d = f_d^1(X_t)$, so $A_{d,1} = \lambda_{d,0}$, $B_{d,1} = \lambda_{d,1}$, $C_{d,1} = H_d$.

Conjecture for $n \geq 2$: $z_{n,t}^d = A_{d,n} + B'_{d,n}X_t + \frac{1}{2}X'_t C_{d,n}X_t$.

Before deriving the recursions it is essential to note a structural difference between the pricing equations for strips and bonds. For a bond, no-arbitrage gives $E_t(\cdot) + \frac{1}{2}\text{Var}_t(\cdot) = 0$, so the variance term enters with a minus sign in the price coefficients. For a strip, the pricing equation is $z_{n,t}^d = E_t(\cdot) + \frac{1}{2}\text{Var}_t(\cdot)$, so the variance enters with a plus sign. The dividend payoff is a convex function of shocks, so strip holders benefit from volatility, while bond holders are penalized. This is why the term $\bar{C}_{d,n-1}\Omega_x\bar{C}_{d,n-1}$ appears with a + sign in (73) below, whereas the analogous terms in (56) and (64) carry a minus sign.

The unexpected part of $\Delta d_{t+1} + m_{t+1} + z_{n-1,t+1}^d$ loads on $(Y_{t+1} - E_t(Y_{t+1}))$ with vector $(h_d - h(X_t) + \tilde{h}_{d,n-1})$, where $h_d = [0, \rho_{cd}, 1, 0]'$ and:

$$\tilde{h}_{d,n-1}(X_t) = \begin{pmatrix} B_{d,n-1} + 2\bar{C}_{d,n-1}X_t \\ 0 \\ 0 \\ 0 \end{pmatrix}.$$

The inflation entry is 0, unlike the nominal bond case, because the strip pays a real dividend and m_{t+1} is the real SDF. The full loading vector is:

$$h_d - h(X_t) + \tilde{h}_{d,n-1}(X_t) = \begin{pmatrix} -(1-\theta)k_{c,2}\Gamma_c(X_t) + B_{d,n-1} + 2\bar{C}_{d,n-1}X_t \\ \rho_{cd} - \gamma \\ 1 \\ 0 \end{pmatrix}. \quad (67)$$

Using the block-diagonal structure of Ω :

$$\frac{1}{2}\text{Var}_t(\cdot) = \frac{1}{2} \left\| -(1-\theta)k_{c,2}\Gamma_c + B_{d,n-1} + 2\bar{C}_{d,n-1}X_t \right\|_{\Omega_x}^2 + \frac{1}{2}(\rho_{cd} - \gamma)^2\sigma_c^2 + \frac{1}{2}\sigma_d^2.$$

Expanding and collecting by degree, the cross-terms and pure-squared terms contribute:

$$\text{degree 0: } (\theta - 1)k_{c,2}\lambda'_{c,1}\Omega_x B_{d,n-1} + \frac{1}{2}B'_{d,n-1}\Omega_x B_{d,n-1}, \quad (68)$$

$$\text{degree 1: } (\theta - 1)k_{c,2} \left[X'_t \bar{H}'_c \Omega_x B_{d,n-1} + B'_{d,n-1} \Omega_x \bar{C}_{d,n-1} X_t \right] + 2B'_{d,n-1} \Omega_x \bar{C}_{d,n-1} X_t, \quad (69)$$

$$\text{degree 2: } 2(\theta - 1)k_{c,2} X'_t \bar{H}'_c \Omega_x \bar{C}_{d,n-1} X_t + 2X'_t \bar{C}_{d,n-1} \Omega_x \bar{C}_{d,n-1} X_t. \quad (70)$$

Applying $z_{n,t}^d = E_t(\cdot) + \frac{1}{2}\text{Var}_t(\cdot)$ and matching by degree:

$$A_{d,n} = A_{d,n-1} + \lambda_{d,0} + \frac{(B_{d,n-1} + 2(\theta - 1)k_{c,2}\lambda_{c,1})'\Omega_x B_{d,n-1}}{2}, \quad (71)$$

$$B_{d,n} = \lambda_{d,1} + \left(I + \Phi + \Omega_x((\theta - 1)k_{c,2}\bar{H}_c + \bar{C}_{d,n-1}) \right)' B_{d,n-1} + (\theta - 1)k_{c,2}\bar{C}_{d,n-1}\Omega_x\lambda_{c,1}, \quad (72)$$

$$C_{d,n} = H_d + C_{d,n-1} + 2\bar{C}_{d,n-1}\Phi + \bar{C}_{d,n-1}\Omega_x\bar{C}_{d,n-1} + 2(\theta - 1)k_{c,2}\bar{H}_c\Omega_x\bar{C}_{d,n-1}. \quad (73)$$

The $+\bar{C}_{d,n-1}\Omega_x\bar{C}_{d,n-1}$ term in (73) contrasts with the $-\bar{C}_{\$,n-1}\Omega_x\bar{C}_{\$,n-1}$ in (56) and $-\bar{C}_{n-1}\Omega_x\bar{C}_{n-1}$ in (64), for the reason explained above.

19.3.2 Equity Risk Premium of the Strip

The one-period expected excess return of the n -period dividend strip over the real short rate is:

$$rp_{n,t}^d = E_t[z_{n-1,t+1}^d - z_{n,t}^d + \Delta d_{t+1} - r_{1,t}].$$

Using the Le-Singleton approximation for $E_t[z_{n-1,t+1}^d]$:

$$E_t[z_{n-1,t+1}^d] \approx A_{d,n-1} + B'_{d,n-1}(I + \Phi)X_t + \frac{1}{2}X'_t C_{d,n-1}X_t + X'_t \bar{C}_{d,n-1}\Phi X_t.$$

Subtracting $z_{n,t}^d = A_{d,n} + B'_{d,n}X_t + \frac{1}{2}X'_t C_{d,n}X_t$, adding (39) (since $E_t[\Delta d_{t+1}] = g_d(X_t)$), and subtracting (58):

$$\begin{aligned} rp_{n,t}^d \approx & \underbrace{A_{d,n-1} - A_{d,n} + g_{d,0} - A_1}_{\text{constant}} + \underbrace{(B_{d,n-1}(I + \Phi) + g_{d,1} - B_{d,n} - B_1)'X_t}_{\text{linear}} \\ & + \underbrace{\frac{X'_t(C_{d,n-1} - C_{d,n} - C_1 + W_d + 2\bar{C}_{d,n-1}\Phi)X_t}{2}}_{\text{quadratic}}. \end{aligned} \quad (74)$$

The difference relative to the bond risk premium (65) is that the linear term here contains $g_{d,1}$ (dividend sensitivity to state variables) instead of a contribution from $g_{\pi,1}$ (inflation sensitivity). Since $g_{d,1}$ and $g_{\pi,1}$ can have opposite signs, the two term structures can slope in opposite directions. This is the mechanism through which the quadratic model resolves the problem documented in Section 14.2

20 Why the Linear Model Fails and How the Quadratic Model Fixes It

20.1 The Standard Calibration Problem

In the one-factor Bansal–Yaron (2004) model, denote expected consumption growth by x_t :

$$\Delta c_{t+1} = \mu_c + x_t + \varepsilon_{c,t+1}, \quad x_{t+1} = \rho_x x_t + \varepsilon_{x,t+1}, \quad \varepsilon_{x,t+1} \sim \mathcal{N}(0, \sigma_x^2), \quad (75)$$

$$\Delta d_{t+1} = \mu_d + q x_t + \varepsilon_{d,t+1}, \quad \varepsilon_{c,t+1} \sim \mathcal{N}(0, 1), \quad \varepsilon_{d,t+1} \sim \mathcal{N}(0, 1). \quad (76)$$

Since the log wealth-consumption ratio is linear ($H_c = 0$), the market price of risk for $\varepsilon_{x,t+1}$ is the constant $(\theta - 1)k_{c,2}\lambda_{c,1}$. With $\lambda_{c,1} = (1 - 1/\psi)/(1 - k_{c,2}\rho_x)$ (from equation 36 with $\bar{H}_c = 0$), the log price-dividend ratio of the strip is $z_{n,t}^d = A_{d,n} + B_{d,n}x_t$ with recursion $B_{d,n} = q - 1/\psi + \rho_x B_{d,n-1}$, $B_{d,0} = 0$, giving:

$$B_{d,n} = \frac{q - 1/\psi}{1 - \rho_x} \cdot \frac{1 - \rho_x^n}{1 - k_{c,2}\rho_x}.$$

(The factor $(1 - k_{c,2}\rho_x)^{-1}$ enters via $\lambda_{c,1}$, not $B_{d,n}$ directly; it appears when computing the risk premium $rp_n^d = (\theta - 1)k_{c,2}\lambda_{c,1} \cdot B_{d,n-1}\sigma_x^2$.) This yields:

$$rp_n^d = \frac{(q - 1/\psi)(1 - \rho_x^n)(\gamma - 1/\psi)k_{c,2}}{(1 - \rho_x)(1 - k_{c,2}\rho_x)}\sigma_x^2. \quad (77)$$

Under standard calibration ($\gamma > 1 > 1/\psi$, $q > 1/\psi$, $\rho_x > 0$): $(q - 1/\psi) > 0$, $(\gamma - 1/\psi) > 0$, and $(1 - \rho_x^n)$ increases in n , so rp_n^d is positive and increasing in n . This result, (Croce et al. 2015) shows the upward-sloping term structure of real bond risk-premia.

The mean real bond risk premium is:

$$rp_n = -\frac{(1 - \rho_x^n)(\gamma - 1/\psi)k_{c,2}}{\psi(1 - \rho_x)(1 - k_{c,2}\rho_x)}\sigma_x^2. \quad (78)$$

Under the same calibration this is negative and decreasing in n . The term structure is downward-sloping.

The Sharpe ratio of dividend strip is:

$$SR_n = \frac{(q - 1/\psi)\left(\frac{1 - \rho_x^n}{1 - \rho_x}\right)\frac{(\gamma - 1/\psi)k_{c,2}\sigma_x^2}{1 - k_{c,2}\rho_x}}{\sqrt{\left(\frac{1 - \rho_x^{n-1}}{1 - \rho_x}\right)^2 (q - 1/\psi)^2 \sigma_x^2 + \sigma_d^2}}. \quad (79)$$

Dividing numerator and denominator by $(1 - \rho_x^{n-1})$, one verifies that SR_n increases in n . Here we have again a upward-sloping term structure.

The problem here is real: the preference for early resolution of uncertainty simultaneously raises the insurance value of long real bonds (depressing their yields and making their risk premia negative) and raises the disutility of long-horizon dividend risk, inflating the premia

on long-duration strips. So, under the fact that market price of risk is constant, we face the problem that fixing one worsens the other.

20.2 The Quadratic Model's Solution

With $H_c \neq 0$, the mean excess returns take the form:

$$rp_n^{\text{real, Q}} = (\theta - 1)k_{c,2} \left[\lambda'_{c,1} \Omega_x B_{n-1} + 2 \cdot \text{tr}(\bar{H}_c \Omega_x \bar{C}_{n-1} \mathbb{E}(X_t X'_t)) \right], \quad (80)$$

$$rp_n^{d,Q} = (1 - \theta)k_{c,2} \left[\lambda'_{c,1} \Omega_x B_{d,n-1} + 2 \cdot \text{tr}(\bar{H}_c \Omega_x \bar{C}_{d,n-1} \mathbb{E}(X_t X'_t)) \right] + \gamma \rho_{cd} \sigma_c^2, \quad (81)$$

$$SR_n = \frac{rp_n^{d,Q}}{\sqrt{B'_{d,n-1} \Omega_x B_{d,n-1} + \text{tr}(\bar{C}_{d,n-1} \Omega_x \bar{C}_{d,n-1} \mathbb{E}(X_t X'_t)) + \sigma_d^2}}. \quad (82)$$

Derivation of (80) is obtained by taking expectations of (65): the linear term in X_t vanishes since $\mathbb{E}(X_t) = 0$; the quadratic term uses $\mathbb{E}[X'_t M X_t] = \text{tr}(M \mathbb{E}(X_t X'_t))$ for any matrix M . This gives:

$$\mathbb{E}(rp_{n,t}) = (A_n - A_{n-1} - A_1) + \frac{1}{2} \text{tr}((C_n - C_{n-1} - C_1 - 2\bar{C}_{n-1}\Phi) \mathbb{E}(X_t X'_t)).$$

Substituting the recursions (62) and (63), the terms in $A_n - A_{n-1} - A_1$ reduce to $(\theta - 1)k_{c,2}\lambda'_{c,1}\Omega_x B_{n-1} - \frac{1}{2}B'_{n-1}\Omega_x B_{n-1}$; the terms from $C_n - C_{n-1} - C_1 - 2\bar{C}_{n-1}\Phi$ reduce via (64) to $-\bar{C}_{n-1}\Omega_x \bar{C}_{n-1} + 2(\theta - 1)k_{c,2}\bar{H}_c \Omega_x \bar{C}_{n-1}$. Taking the trace expectation and rearranging, the $B'_{n-1}\Omega_x B_{n-1}$ terms cancel and (80) follows. The derivation of 81 proceeds identically, replacing the real bond recursions 62- 64 with the strip . 82 follows directly from the ratio of 81 to the conditional standard deviation of the strip return. The term $\gamma \rho_{cd} \sigma_c^2$ in (81) survives because it represents the direct covariance between dividend and consumption shocks (present regardless of curvature).

The additional term $2 \cdot \text{tr}(\bar{H}_c \Omega_x \bar{C}_{n-1} \mathbb{E}(X_t X'_t))$ in (80) and (81) exists only because $\bar{H}_c \neq 0$. Because $\lambda_{c,1}$ and $\bar{H}_c$ can be chosen independently of γ and ψ , the model can independently calibrate:

- the slope of the real bond term structure through $\bar{H}_c$ in (80);
- the slope of the equity term structure through $\bar{H}_d$ (which drives $\bar{C}_{d,n-1}$) in (81);
- the slope of the Sharpe ratio term structure through both in (82).

21 Relationship to the Linear Model and Parameter Restrictions

Setting $H_c = H_d = 0$ recovers the standard linear long-run risks model as a special case. All terms in W_c, W_d, W_π contain H_c or $\bar{H}_c$, so $W_c = W_d = W_\pi = 0$: expected cash flows become linear in X_t and risk premia are constant across maturities.

The free parameters then admit a natural economic interpretation. With $H_c = 0$, 36 gives $g_{c,1} = [k_{c,1}I - k_{c,2}\Phi]' \lambda_{c,1} / (1 - 1/\psi)$. Requiring $g_{c,1} = (1, 0)'$, so that $\mathbb{E}_t(\Delta c_{t+1}) = g_{c,0} + x_{1,t}$, uniquely pins down:

$$\lambda_{c,11} = \frac{1 - 1/\psi}{k_{c,1} - k_{c,2}\phi_{11}}, \quad \lambda_{c,12} = \frac{k_{c,2}\phi_{12}(1 - 1/\psi)}{(k_{c,1} - k_{c,2}\phi_{22})(k_{c,1} - k_{c,2}\phi_{11})}. \quad (83)$$

The denominator in $\lambda_{c,11}$ is the $(1, 1)$ entry of $(k_{c,1}I - k_{c,2}\Phi)$, and the full system (83) can be written as $\lambda_{c,1} = (1 - 1/\psi)[(k_{c,1}I - k_{c,2}\Phi)^{-1}]' e_1$, provided $\det(k_{c,1}I - k_{c,2}\Phi) \neq 0$. This identifies $x_{1,t}$ as expected consumption growth.

Setting $\alpha_c = 1/\psi$ in (45) eliminates the $g_{c,1}$ term, leaving:

$$g_{\pi,1} = \frac{i_m}{1 - \alpha_\pi} = \frac{(0, 1)'}{1 - \alpha_\pi},$$

so $\mathbb{E}_t(\pi_{t+1}) = g_{\pi,0} + x_{2,t}/(1 - \alpha_\pi)$: the second state variable drives expected inflation directly, identifying $x_{2,t}$ as the inflation factor.

Finally, with $\bar{H}_c = 0$ and $\alpha_c = 1/\psi$, using $\theta(1 - 1/\psi) = 1 - \gamma$:

$$g_{d,1} = \lambda_{d,1} + [\gamma + (\theta - 1)(1 - 1/\psi)] g_{c,1} = \lambda_{d,1} + \frac{g_{c,1}}{\psi}.$$

If $\lambda_{d,12} = 0$ then $g_{d,1} = (\lambda_{d,11} + 1/\psi)g_{c,1} \equiv q \cdot g_{c,1}$, recovering the constant leverage ratio of Bansal and Yaron (2004).

22 Conclusion

We have presented the complete analytical derivation of Doh and Wu (2017). The central idea is that allowing the log -wealth consumption ratio to be a quadratic function of the state vector X_t , following Le and Singleton (2010), makes time-varying market prices of risk. Combined with the reverse-engineering approach to expected cash flows, this breaks the structural link that prevents the standard Bansal-Yaron (2004) calibration from simultaneously matching the term structures of bond and equity risk premia.

The result is a model that jointly achieves an upward-sloping term structure of bond risk premia, a downward-sloping term structure of equity risk premia, and a downward-sloping term structure of Sharpe ratios on dividend strips. These are three empirical patterns identified in van Binsbergen et al. (2012) and Beeler and Campbell (2012) that the linear model cannot reconcile. Crucially, all three obtain under the same preference calibration, without abandoning the preference for early resolution of uncertainty that makes the long-run risks framework work in first place.

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